Experimental apparatus for universal spin-motion control in multi-ion registers

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k-copy Randomised Benchmarking

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To construct useful scale quantum processors we must house many qubits physically close together, and must ensure the full register operates with "low error rates". But how should we define our errors in a large system? Naively, one might hope that the error rate of a multi-qubit system could be inferred from the error rates of its constituent qubits, but unfortunately, not all errors are commensurable — there are a zoo of possible error channels. They may be coherent, incoherent, continuous, stochastic, local, correlated, time dependent or independent (and I am sure a host of further adjectives exist). In response, a zoo of error correction and mitigation techniques have been developed each with their own scaling overheads and assumptions on these noise models they treat. It is a large and intimidating search space, and so, to better equip ourselves for navigation, we require tools to characterise what errors are relevant in our physical systems. In this chapter we introduce a new tool for characterising correlated errors in multi-qubit systems: *k-copy randomised benchmarking*. This technique extends single-qubit Clifford RB (discussed in chapter **??**) to parallel application across multiple qubits, where now symmetry resolved dynamics provides evidence of, and quantifies various multi-qubit error channels.

This chapter is structured as follows: we start by giving an overview on the problem of *characterisation*, and what techniques already exist to address it. We then outline the k-copy protocol and discuss the relevant symmetric subspace structure. We derive the population dynamics when subject to simple noise models and discuss variations of the protocol that isolate specific subspaces. We provide experimental demonstrations of the protocol on a trapped-ion processor through injecting known correlated and uncorrelated errors. We end this chapter by demonstrating the scalability of the protocol on an 8-qubit register, and discuss the limitations and potential of this method for further scaling to larger systems.

1.1 Errors and Characterisation

Errors, which we shall give a broad definition as any deviations between the realised and ideal state of a quantum information processor (QIP), are due to unintentional environmental coupling and control imperfections. They are unavoidable, and if not carefully considered or mitigated, lead to unreliable or unverified computation. The process of modelling and quantifying these errors is known as *characterisation*. Characterising errors has two practical advantages: 1) for the engineer, quantifying an error allows for a clear signal to improve hardware and calibration routines, and 2) for the theorist, understanding physically motivated error channels allows for the development of error correction and mitigation techniques that are robust to real world noise. Both users are concerned ultimately with the same question: How faithfully can we execute arbitrary circuits on our physical devices? Extracting metrics to this end is a wide research area in quantum information, with techniques of varying scalability and implicit assumptions on underlying noise models [1]. On one end of the spectrum sits full process tomography**citation** which reconstructs quantum channels, here represented as completely positive, trace preserving (CPTP) maps on the space of density matrices,

$$\mathcal{E} : \mathbb{C}^{d\times d} \rightarrow \mathbb{C}^{d\times d}, \tag{1.1}$$

where the dimension of the Hilbert space is $d = 2^n$ for n qubits. The CPTP map is rich in information, capturing many relevant error channels, but has poor scaling with system size, generally required $O(d^4)$ parameters to describe, and $O(d^4)$ measurements to reconstruct. This scaling fundamentally limits tomography to small system sizes. The goal then shifts to how to extract useful information in predicting the behaviour of a system executing a circuit, without full channel reconstruction. Now, two strategies emerge: 1) we make assumptions on the underlying noise model, or 2) we actively simplify the structure of the error channel through averaging. The first strategy leads to techniques such as noise learning **citation** and the latter leads to random circuit methods, such as randomised benchmarking (RB) [**magesan_characterizing_2012**, 2, 3]. Both approached dramatically reduce the number of parameters required to describe the system, and the number of measurements required to extract them.

The assumptions on the noise model are cleanly motivated by considering the modularity of quantum circuits, and how errors may violate this modularity.

Following the description of imperfect computation and crosstalk errors given in Refs. [4], and [5], respectively, we define the following models of QIPs, depicted graphically in figure **??**. Panel a) shows with the most desirable, yet technically unfeasible *Ideal QIP*, described by a *closed quantum system*. Here, the qubits comprise a quantum state represented by a single vector in Hilbert space. The state is isolated from the environment and may be decomposed into a collection of subsystems that remain uncoupled unless explicit multi-qubit operations are applied. Operations are all unitary evolutions, conveniently described by single and two-qubit unitaries, which we term *gates*. Errors may exist, but are limited to reversible unitary interactions. In b) we introduce external and control field noise due to an environment with no history, i.e. a source of Markovian noise. We must now depart from the quantum state vector and unitary evolution description, and instead use density matrices and CPTP maps to describe the state and dynamics of the QIP. We highlight a subset of these models, termed *Crosstalk-Free Markovian QIP*, where the dynamics of distinct subsystems are governed by locally acting CPTP maps. To satisfy this condition, operations are said to be *local* and *independent* [5] — *local* is the assumption that only explicitly applied multi-body operations lead to correlations between qubits that participate in the ideal multi-body operation; *independent* is the assumption that the action of an applied gate on a subset of qubits is independent of any other gates applied in parallel on the other qubits. With these two criteria satisfied, we may retain the composibility of gates, conveniently allowing us good prediction of the behaviour of a many-qubit deep circuit, using only characterisations of each gate in a universal gate set applied to small subsystems. If these criteria are not met, to accurately predict the behaviour of a deep circuit, each unique layer must be characterised, which is intractable.

The last class of models, included for completeness, are *Non-Markovian QIP*, where we relax the assumption that the effect of the environment has no history dependence. In this case, any instantaneous state may still be able to be described by a density matrix, however the dynamics can no longer be described by the application of a sequence of CPTP maps. To accurately characterise the system, full process tomography would need to be implemented for each possible circuit.

Practically, to ensure we can reasonably characterise our physical systems, and predict their performance on novel circuits, we must design and appropriately isolate hardware to be well approximated by the *Crosstalk-Free Markovian* models. The next section provides some background on existing techniques to verify crosstalk-free performance, and provide some quantification when the assumption is violated.

1.2 Existing approaches to correlated-noise characterisation

Existing approaches to correlated-noise characterisation span a trade-off between expressiveness and scalability. The most expressive, and arguably least scalable, are tomographic techniques, such as gate set tomography [**nielsen_gate_2021**] or the crosstalk sensitive variation [6]. Here, the implicit assumption that the noise environment is Markovian is sufficient, and no other assumptions of underlying noise models are required. These techniques are valuable in producing detailed CPTP maps that inform not only the presence of crosstalk-free violating noise, but also offer evidence on the underlying noise process. Demonstrations are however limited to small system sizes of two-qubits in [6].

Noise learning protocols are substantially more scalable by imposing a restricted set of noise models, and fitting experimental data to these models. Correlated-noise specific variations of this approach [7, 8] apply repeated random circuits that act to simplify the action of noise channels to Pauli type errors, and then through measurements and stastitical inference, likely Pauli error models are found, and crosstalk-free violating errors may be quantified. Noise learning can also be adaptable by first fitting simple models, for example when twirling over random circuits yields structured error channels, and progressively increasing the complexity of the inferred model when arbitrary circuits are run on the same device [9]. This nested-model approach provides a framework for robust characterisation of the system, however, demonstrations are still currently limited to small systems.

Similar to enforcing Pauli-type errors, randomised benchmarking techniques average the action of noise channels to impose structure to general error channels. This is achieved by twirling the action of the channel over deep circuits comprised of a chosen gate group. Clifford twirling techniques, notably the standard RB protocol [2, 10], have been adapted for detecting crosstalk induced correlations by simultaneous RB [11], and correlated RB [12] protocols. Correlated RB uses independent local twirling of multiple system copies to extract multi-qubit survival decay rates with Clifford sequence length. The weight and locality of correlated errors may then be inferred through a distance of the measured channel from a product (uncorrelated) channel. Other, non-Clifford twirled RB protocols, such as cycle-benchmarking [3] and EPLG [13] apply parallel gate sequences on multiple subsystems to extract some error applied per layer, that can then be used to infer error contributions due to correlated-noise channels. These methods have been shown to scale to 10s of qubits, and provide system level diagnostics of device performance.

We note two further streams of current research, first the application of RB in determining the unitarity of noise [14], and the extension of this work to quantifying bipartite unitary noise [**girling_estimation_2022**], which can induce correlations across the qubit register. Second, the use of custom circuits designed to probe crosstalk-free violations [**rudinger_probing_2019**, 5, 15]. **expand section**.

The present protocol we have developed, and demonstrated experimentally, sits firmly in the RB methodology. It however differs from simultaneous and correlated RB by providing not only a quantification of crosstalk-free violating noise, but also providing evidence for the action of the error channel on a multi-qubit system. This extra information is inferred through direct observation of population decay and coupling between the subspaces enforced by twirling over the diagonally applied single-qubit Clifford group. The inspiration of this work comes from correlation spectroscopy [16] and subspace-leakage techniques [17, 18]. Correlation spectroscopy demonstrates that common-mode noise can be revealed through its action on collective and differential degrees of freedom, while subspace-leakage approaches provide a framework for defining error models not by full CPTP maps, but rather by their action on a discrete set of physically relevant subspaces. In such settings,

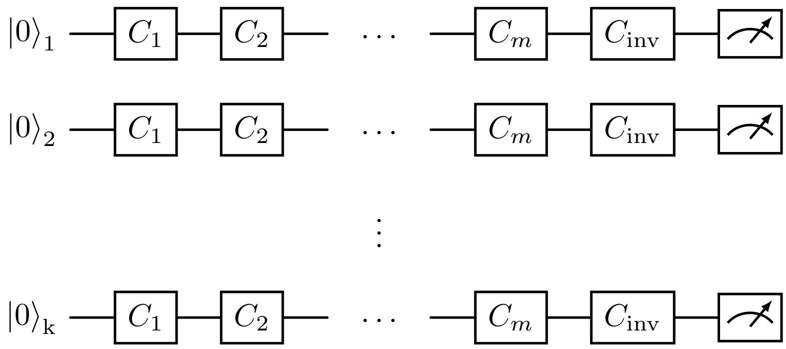

Figure 1.1: Circuit diagram for k-copy RB.

symmetry is not merely a simplifying assumption providing structure to arbitrary channels, but rather it is the very mechanism that separates crosstalk-free violating errors from local noise. The next section details the subspaces enforced by diagonally applied single-qubit Clifford twirling, and provides the protocol for *k-copy* RB.

1.3 The k-copy RB scheme

1.3.1 Protocol

The k-copy RB protocol is implemented through standard single-qubit Clifford RB sequence (see section **??**) applied simultaneously to multiple qubits, with the same randomly sampled Clifford applied to each qubit at every step, as shown in Fig. 1.1. A single sequence is composed of $m-1$ parallel gates sampled randomly from the Clifford group. As in standard RB, the final applied layer is an inverting Clifford, which, due to the Clifford group being closed, may be selected to take the final state back to the initial computational eigenstate. The full sequence of random Clifford gates is then equivalent to the identity operation with some unknown noise overlaid[1]. Practically, each gate layer can be executed using a global field, or

[1]Note: To further symmetrise state preparation and measurement errors, we may allow the final target state to be randomised between $|0\rangle$ and $|1\rangle$ for each sequence [10].

through parallel local control lines addressing each qubit. A final $\langle Z \rangle$ measurement is made in parallel on all qubits to probe the successful execution of the Clifford sequence. Qubits measured in the target state are labelled $\mathcal{S}$, and qubits measured in the opposite state are labelled $\mathcal{F}$, for success and failure respectively. To build statistics of success and failure rates, r random circuits are trialled for each length m, and each circuit is repeated s times to reduce shot noise. To retain information on multi-qubit correlations, for every experimental shot, data from each qubit pair (i, j) is assigned to one of four outcomes, $\mathcal{P}_{SS}^{(i,j)}$, $\mathcal{P}_{SF}^{(i,j)}$, $\mathcal{P}_{FS}^{(i,j)}$, or $\mathcal{P}_{FF}^{(i,j)}$, where S and F denote success and failure. As with other RB protocols, the success probabilities versus sequence length are then fit to explicitly defined decay curves inferred by the action of the twirled error models.

By limiting the analysis to pairwise comparisons, the *k-copy* protocol is both efficient in the number of trials required on the device, and in the postprocessing of the recovered data. For a register of k qubits, the same shot record contains outcomes for all $k(k-1)/2$ pairs. The two-body error models are derived in the following sections for prevalent noise mechanisms in both ion trapping and superconducting information processors. Higher weight correlations are discussed further in section 1.5.

1.3.2 k-copy Clifford analysis

This section develops the representation-theoretic framework required to analyze the decay curves arising in the k-copy RB protocol. We begin by considering the representations of the single-qubit Clifford group acting diagonally on two copies, $Cl_1^{\otimes 2}$ (see Ref. [19] for in detail analysis). The irreducible representations give a natural (Schur) basis to consider how possible error channels are simplified upon the k-copy RB protocol. We finish this subsection by deriving the correlated populations observed experimentally when the system is subject to common random-unitary single-qubit noise.

Representations of the 2-copy Clifford group

We consider a group $\mathcal{G}$ and a representation $r : \mathcal{G} \to GL(\mathcal{L})$, where GL is the general linear group (consisting of the set of invertible matrices and the matrix multiplication operation), and $\mathcal{L}$ denotes the Liouville (vectorised operator) space of two qubits. Twirling over the elements g in $\mathcal{G}$ enforces symmetry on a general error channel, Λ, given in Liouville form,

$$\tilde{\Lambda} = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} r(g)\Lambda r(g)^{\dagger}, \tag{1.2}$$

where $|\mathcal{G}|$ is the number of elements comprising the finite group $\mathcal{G}$. The twirling enforces commutation with the group action,

$$r(g)\tilde{\Lambda}r(g)^{\dagger} = \tilde{\Lambda} \quad \forall g \in \mathcal{G}. \tag{1.3}$$

The representation of the space, $\mathcal{L}$ decomposes to irreps λ as,

$$\mathcal{L} = \bigoplus_{\lambda}(V_\lambda \otimes \mathbb{C}^{m_\lambda}), \tag{1.4}$$

where V_λ is the irreducible representation space, and $\mathbb{C}^{m_\lambda}$ is a complex valued multiplicity space with dimension of the multiplicity m_λ. From Schur's lemma, Eq. 1.3 implies the twirled error channel has the form,

$$\tilde{\Lambda} = \bigoplus_{\lambda}(I_\lambda \otimes \mathcal{A}_\lambda), \tag{1.5}$$

where $\mathcal{A}_\lambda$ is the multiplicity matrix of dimensions $m_\lambda \times m_\lambda$.

The 2-copy single-qubit Clifford group, analysed in detail in references [19, 20], decomposes the general 16-dimensional two qubit space into 4 irreducible representations, irreps, with dimensions $\{1, 2, 3, 3'\}$, (the notation $3'$ is used to contrast the other $3-$dimensional irrep), and multiplicities $\{2, 1, 3, 1\}$. The sub-representations are summarised in table 1.1. The irreps are uniquely determined by their character on the conjugacy classes of the Clifford group. Two group elements $g, h \in G$ are said to be in the same conjugacy class if they are related by conjugation,

$$g = khk^{-1}, \tag{1.6}$$

Sub-representation	irrep	Basis operators
Identity	1	II
Left qubit	3	XI, YI, ZI
Right qubit	3	IX, IY, IZ
Trace	1	$(XX + YY + ZZ)/\sqrt{3}$
Diagonal traceless	2	$(XX + YY - 2ZZ)/\sqrt{6}$, $(XX - YY)/\sqrt{2}$
Symmetric off-diagonal	$3'$	$(XY + YX)/\sqrt{2}$, $(XZ + ZX)/\sqrt{2}$, $(YZ + ZY)/\sqrt{2}$
Antisymmetric	3	$(XY - YX)/\sqrt{2}$, $(XZ - ZX)/\sqrt{2}$, $(YZ - ZY)/\sqrt{2}$

Table 1.1: Sub-representations of the $Cl_1^{\otimes 2}$ group. The irrep, dimension of each sub-representation, and a set of operators in the Pauli basis which span the subspace dimension is given. The irrep basis vectors are from Eq. 56 in reference [20].

Class	Representation	Member
Identity	e	I
Swap	(12)	H
Two-swap	(12)(34)	X
Three-cycle	(123)	HS
Four-cycle	(1234)	S

Table 1.2: Conjugacy classes for the single qubit Clifford group.

irrep	e	(12)	(12)(34)	(123)	(1234)
Identity/Trace	1	1	1	1	1
Left/Right/Antisymmetric	3	-1	-1	0	1
Diagonal traceless	2	0	2	-1	0
Symmetric off-diagonal	3	1	-1	0	-1

Table 1.3: Character table for the Clifford group given the irreps in table 1.1, and with conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle. The "sign" irrep of the single-qubit Clifford group has zero multiplicity in the two-copy case, and so is excluded here.

where $k \in G$. With this definition, the 24 single qubit Clifford elements are divided into five permutation classes, shown in table 1.2, with a representative name and example member following each class. These classes make obvious the known isomorphism between Cl_1 and the symmetric group S_4 (modulo phase). Character tables [**lee_character_2023**], are useful tools in representation theory, to uniquely describe irreps by the traces of the representative matrices for group elements of each conjugacy class. The table is presented as a 2D matrix with rows corresponding to group representations and columns to group conjugacy classes. The character, ξ, is defined by,

$$\xi_\lambda(g) = \mathrm{Tr}[r_\lambda(g)], \tag{1.7}$$

where r_λ is the irreducible component of the representation of G acting on some vector space,

$$r_\lambda(g)_{i,j} = \mathrm{Tr}[B_i^\dagger g B_j g^\dagger], \tag{1.8}$$

where B_i are the orthonormal basis vectors for irrep λ. Taking the two-qubit subrepresentations reproduced in table 1.1, the character table calculated is shown in table 1.3. We can see confirmed that the left, right, and antisymmetric subrepresentations form a multiplicity of the $3D$-irrep labelled 3, while the trace subrepresentation is a multiplicity of the $1D$ trivial irrep.

Given the structure of a general twirled map is enforced to be block diagonal in the irrep basis, given by Eq. 1.5, we can do a simple count to find generally how many parameters are required to fully describe the CPTP map. Accounting for the multiplicities, and the constraint that the map is trace preserving, we find that at most $2^2 + 3^2 + 1 + 1 - 2 = 13$ independent scalars are needed to describe $\tilde{\Lambda}$. The diagonal terms, here labelled $\vec{\alpha} = (\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as})$, describe a decay rate contribution from each sub-representation, whilst off-diagonal terms, $\vec{m} = (m_{id\to tr}, m_{L\to R}, m_{R\to L}, m_{L\to as}, m_{R\to as}, m_{as\to L}, m_{as\to R})$ describe coupling between the multiplicity spaces.

In practice, different physical error mechanisms populate only a subset of the

Coupling	irrep	Mechanism
$m_{id\to tr}$	1	Decays (non-unital)
$m_{L\to R}$, $m_{R\to L}$	3	Qubit Swap, Correlated decays (non-unital)
$m_{L\to as}$, $m_{R\to as}$, $m_{as\to L}$, $m_{as\to R}$	3	Entanglement

Table 1.4: Possible mechanisms that result in the coupling of sub-representation sectors of the $Cl_1^{\otimes 2}$ twirled error channel.

coefficients α and m, leading to simplified dynamics with characteristic signatures of the underlying process. Table 1.4 provides possible physical mechanisms capable of coupling isotypic representation subspaces. We shall devote some time in the next two sections to a selection of error mechanisms, the first two containing no off-diagonal terms, which greatly simplifies the analytic description of the channel.

1.3.3 Common random unitary noise

It can be seen that the error channels that lead to couplings neatly discern the physical origin of the noise channel. For example, in transmon superconducting qubit platforms, always on ZZ-couplings are a dominant crosstalk error source [**fors_comprehensive_2024**, **alghadeer_crosstalk_2025**, 21], and so the couplings between single qubit subspaces and the antisymmetric subspace can be quantified; in the neutral atom platform, correlated non-unital errors may be observed if the physical qubit separation is of similar order to the qubit transition wavelength [22]; however, in ion trap platforms, environmental noise is often dominated by magnetic and electric field fluctuations, which may act globally on all qubits if they are stored in the same trapping potential. These field fluctuations induce both common and independent single-qubit rotations, but do not couple irrep subspaces together (i.e $m = 0$ for $m \in \vec{m}$). Thus, the general error map of this type is now parametrised by at most the 6 scalars, $\vec{\alpha}$, and results in Liouville form diagonal in the Schur basis, $\mathcal{S}$, (ordered by table 1.1),

$$\tilde{\Lambda}^{\mathcal{S}} = diag\left(I_1, \alpha_L I_3, \alpha_R I_3, \alpha_{tr} I_1, \alpha_{dt} I_2, \alpha_{so} I_3, \alpha_{as} I_3\right), \tag{1.9}$$

where I_d is the d-dimensional identity. Specifically, we consider the effect of noise due to common small-angle single-qubit rotations acting on two-qubits after each gate comprising the desired circuit. The general form for this error channel is given in main text Eq. **??**. A subset of error channels following this description cannot be described by the separable map,

$$\mathcal{E}(\rho_1 \otimes \rho_2) = \mathcal{E}_1(\rho_1) \otimes \mathcal{E}_2(\rho_2), \tag{1.10}$$

where ρ_i is the density matrix of qubit i, as such, the channel induces correlations between qubits in a register. The single qubit rotation is parametrised by rotation angle, θ, and the axis of rotation $\vec{n}$.

$$U(\theta, \hat{n}) = e^{-i\frac{\theta}{2}\hat{n}\cdot\hat{\sigma}}. \tag{1.11}$$

The channel comprised of unitary rotations U_a,

$$\mathcal{E}_a(\rho) = (U_a \otimes U_a)\rho(U_a \otimes U_a)^\dagger, \tag{1.12}$$

is twirled over the discrete group $\mathcal{G}$, with twirling defined by Eq. 1.2, leading to the simplified channel in Liouville form $\tilde{\Lambda}_a$. Considering the form of Eq. **??**, the ensemble of random unitary channels is,

$$\tilde{\Lambda} = \sum_{a\in\mathcal{A}} p_a \tilde{\Lambda}_a. \tag{1.13}$$

Add IRO alternative derivation The goal is now to find the scalar parameters $\vec{\alpha}$, which describe this channel. As discussed in the main text, it is apparent that only the "symmetric trace" sector is preserved under global single-qubit rotations, $\alpha_{tr} = 1$. It is in fact this preservation that leads to the decay into the symmetric state given in the main text Eq. 1.32. The other scalar parameters can be calculated "mechanically" by constructing the Pauli-transfer matrix (PTM) representation and twirling over $Cl_1^{\otimes 2}$ explicitly. However, we note that the implicit symmetry of each sub-representation can be used to define tensor contractions acting on the

$SO(3)^{\otimes 2}$ space spanned by two rotations,

$$\begin{aligned} s_{tr} &= \sum_{i,j,k,l} \delta_{ij}\delta_{kl} A_{ki} B_{ij} = Tr(A^T B), \\ s_{id} &= \sum_{i,j,k,l} \delta_{ik}\delta_{jl} A_{ki} B_{ij} = Tr(A) Tr(B), \\ s_{swap} &= \sum_{i,j,k,l} \delta_{il}\delta_{jk} A_{ki} B_{ij} = Tr(AB), \\ s_{diag} &= \sum_{i,j,k,l} (\delta_{ik}\delta_{jl})\delta_{ij} A_{ki} B_{ij} = \sum_i (A_{ii} B_{ii}), \end{aligned} \tag{1.14}$$

where A and B are the $SO(3)$ representation of independent unitaries acting on the left and right qubits respectively. The scalar parameters $\vec{\alpha}$ are then found by,

$$\begin{aligned} \alpha_{tr} &= \frac{1}{3} s_{tr}, \\ \alpha_{dt} &= \frac{1}{2} s_{diag} - \frac{1}{6} s_{tr}, \\ \alpha_{so} &= \frac{1}{6}(s_{id} + s_{swap}) - \frac{1}{3} s_{diag}, \\ \alpha_{as} &= \frac{1}{6}(s_{id} - s_{swap}). \end{aligned} \tag{1.15}$$

In the case of a global rotation, $A = B$, we find,

$$\begin{aligned} &\alpha_L = \alpha_R = \alpha_{as} = \frac{1 + 2\cos\theta}{3}, \\ &\alpha_{tr} = 1, \\ &\alpha_{dt} = \frac{1}{2}\Big(1 + \cos(2\theta) + 8n_{an}\sin(\theta/2)^4\Big), \\ &\alpha_{so} = \frac{1}{3}\Big(2\cos(\theta) + \cos(2\theta) - 8n_{an}\sin(\theta/2)^4\Big), \end{aligned} \tag{1.16}$$

with an anisotropic term, $n_{an} = n_y^4 + n_z^4 + n_y^2 n_z^2 - n_y^2 - n_z^2$. Taking these scalar products to second order in error angle θ, we find that the anisotropic term may be ignored for small errors,

$$\begin{aligned} \alpha_L &= \alpha_R = \alpha_{as} = 1 - \theta^2/3, \\ \alpha_{tr} &= 1, \\ \alpha_{dt} &= \alpha_{so} = 1 - \theta^2. \end{aligned} \tag{1.17}$$

As such, the dynamics are entirely described by the error angle magnitude. The ensemble of twirled common unitaries across some arbitrary distribution, $\mathcal{A}$, given in Eq. **??**, thus combine to give scalar parameters describing the contribution from each subspace to the observed decay,

$$\alpha_i^{(\mathcal{A})} = \sum_{a\in\mathcal{A}} p_a \alpha_a. \tag{1.18}$$

In the main text, this definition leads to a convenient (and conforming to previous RB work) definition for the correlated error per Clifford given by Eq. **??**, which is the single qubit error contribution to first order for small per-step rotation angles. Comparing to Eq. 1.9, the general ensemble of common single-qubit rotations channel, after twirling over $Cl_1^{\otimes 2}$, is given by the diagonal matrix in the Schur basis,

$$\begin{aligned}\tilde{\Lambda}^{\mathcal{S}}_{U\otimes U} = diag\Big(I_1,\ (1-2\varepsilon_{corr})I_6,\ I_1,\\ (1-6\varepsilon_{corr})I_2,\ (1-2\varepsilon_{corr})I_6\Big).\end{aligned} \tag{1.19}$$

1.3.4 Local depolarising noise

Ultimately, environmental noise will not perfectly couple symmetrically to both qubits, and so independent local single-qubit error channels must also be considered. Assuming an independent noise channel acting on each qubit, Clifford twirling leads to a fully depolarising noise channel of the form,

$$\mathcal{E}_{dp,a}(\rho^{(1q)}) = (1-2\varepsilon_a)\rho^{(1q)} + \varepsilon_a I_2 Tr(\rho^{(1q)}), \tag{1.20}$$

where the superscript denotes that this is for a single qubit subspace, and the subscript a signifies which qubit we are considering. For the “left" and “right" qubits the decay rates are parametrised by ε_L and ε_R, respectively. Comparing to Eq. 1.9, the independent depolarisation channel may be expressed in Liouville form as the diagonal matrix in the Pauli basis,

$$\Lambda^{\mathcal{P}}_{dp} = diag\Big(1, (1-2\varepsilon_L)I_3, (1-2\varepsilon_R)I_3, (1-2\varepsilon_L)(1-2\varepsilon_R)I_9\Big), \tag{1.21}$$

where superscript $\mathcal{P}$ represents the block diagonal matrix is ordered by the two-qubit Pauli basis $\mathcal{P} = \{I\otimes I, \vec{\sigma}\otimes I, I\otimes\vec{\sigma}, \vec{\sigma}\otimes\vec{\sigma}\}$, with $\vec{\sigma} = \{X, Y, Z\}$.

1.3.5 Combined model

Combining the depolarisation channels with the twirled common single-qubit rotation channel,

$$\Lambda_T(\varepsilon_L, \varepsilon_R, \varepsilon_{corr}) = \Lambda_{dp}(\varepsilon_L, \varepsilon_R)\tilde{\Lambda}_{corr}(\varepsilon_{corr}), \tag{1.22}$$

we find scalar parameters,

$$\begin{aligned}
\alpha_L &= 1 - 2\left(\varepsilon_L + \varepsilon_{corr}\right), \\
\alpha_R &= 1 - 2\left(\varepsilon_R + \varepsilon_{corr}\right), \\
\alpha_{tr} &= 1 - 2\left(\varepsilon_L + \varepsilon_R\right), \\
\alpha_{dt} &= \alpha_{so} = 1 - 2\left(\varepsilon_L + \varepsilon_R + 3\varepsilon_{corr}\right), \\
\alpha_{as} &= 1 - 2\left(\varepsilon_L + \varepsilon_R + \varepsilon_{corr}\right),
\end{aligned} \tag{1.23}$$

which fully describe the error channels considered in the main text.

To reproduce the observed multi-exponential decays, we must consider an initial prepared state, and the measurement basis. For the proposed RB experiment, the initial state is always prepared as $|00\rangle$, and the final correlated populations are measured in the computational eigenstates, $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. In the main text, we consider “success" and “failure" rather than “0", and “1", as we allow for target state randomisation, however this detail is excluded here, and we assume that the noise-less RB sequence results in the identity operation. The correlated populations may be given analytically using the derived Liouville operators, and by the vectorized initial and final states,

$$\mathcal{P}_{LR} = \langle\langle L, R|\Lambda_T^m|00\rangle\rangle, \tag{1.24}$$

where L, R denote the successful recovery of the target final state for the “left" and “right" qubit (either $|0\rangle$ or $|1\rangle$) respectively, and m is the number of Clifford gates in the applied sequence. Using this equation, and the explicit transfer matrices, the final populations decays are found,

$$
\begin{aligned}
\mathcal{P}_{SS} &= \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \\
\mathcal{P}_{SF} &= \frac{1}{4} + \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \\
\mathcal{P}_{SF} &= \frac{1}{4} - \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \\
\mathcal{P}_{FF} &= \frac{1}{4} - \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m.
\end{aligned}
\tag{1.25}
$$

Considering the final state of a single qubit after k-copy RB, we find the expected single exponential decay,

$$\mathcal{P}_S^{(L)} = \mathcal{P}_{SS} + \mathcal{P}_{SF} = \frac{1}{2}\left(1 + (1 - 2\left(\varepsilon_L + \varepsilon_{corr}\right)^m\right), \tag{1.26}$$

where the superscript (L) refers to considering the left qubit only. Comparing to standard single qubit RB decay errors parametrised by,

$$\mathcal{P}_S = \frac{1}{2}\left(1 + (1 - 2\varepsilon_c)^m\right), \tag{1.27}$$

where ε_c is the “error-per-Clifford", we find the observed decay of each qubit is now separable into a correlated and independent error component,

$$\varepsilon_c^{(L)} = \varepsilon_L + \varepsilon_{corr}. \tag{1.28}$$

Schematic explanation

The distinction between correlated and uncorrelated noise is illustrated schematically by considering a register in which qubits $i = 3$ and $i = 4$ are dominated by a common rotation error while other qubits are dominated by independent noise. All qubits can exhibit identical single-qubit RB error rates (i.e. an identical depolarised Bloch vector under standard RB), while possessing entirely different pairwise correlation structure: independent noise produces uncorrelated sequence-to-sequence deviations, whereas common noise produces shared deviations between qubits acted on by the same random Clifford sequence. The identical-sequence condition is therefore essential to the protocol: it preserves exactly the symmetry information that would

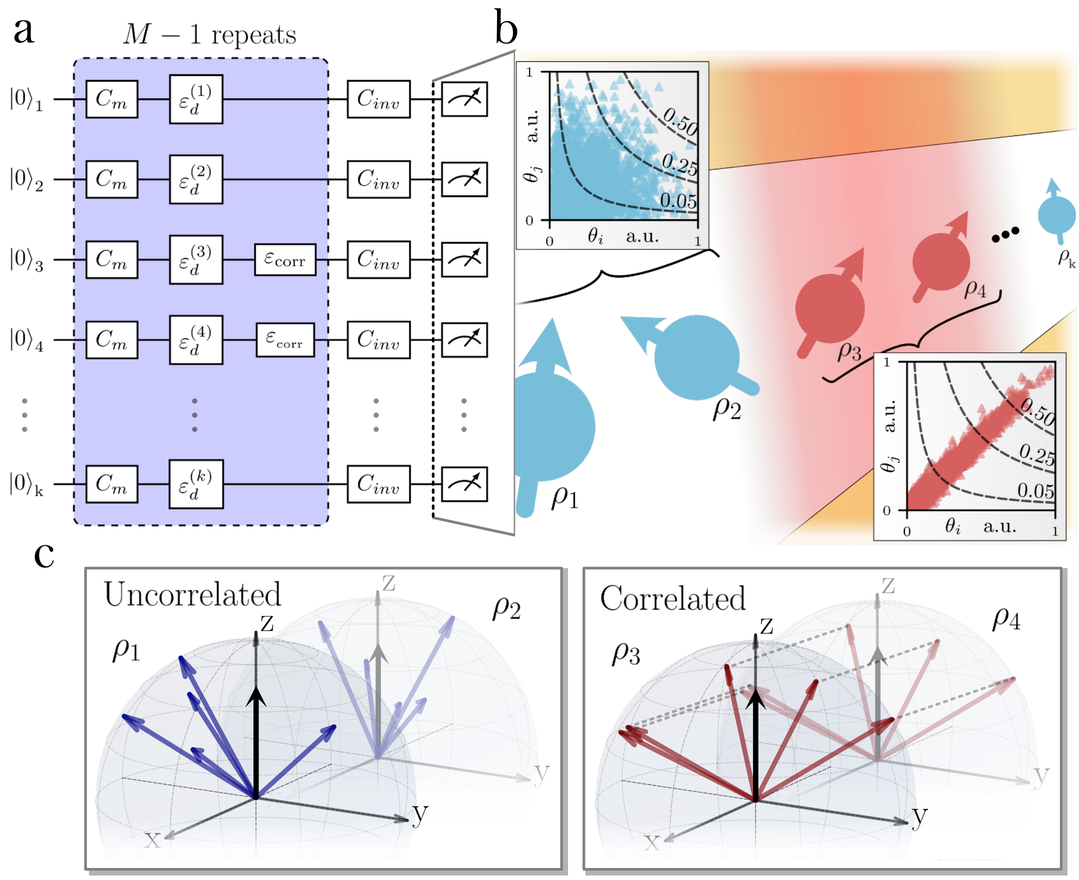

Figure 1.2: **Bloch sphere schematic of corr vs uncorr noise. Include schematic of ion chain prev fig 1c.**

be averaged away under independently sampled local Clifford sequences (as in Simultaneous/Correlated RB, Section 1.2).

This is currently a qualitative, schematic argument (matching the level of detail in the PRL). A thesis chapter could usefully add the joint-angle-distribution analysis that was drafted but cut from the manuscript: uncorrelated noise exponentially suppresses higher-weight error terms $p_w = \prod_i p_i$ relative to single-qubit rates, whereas correlated noise violates this independence assumption and can scale linearly with weight w in the worst case, with consequences for QEC thresholds assuming independent errors ($p_t \propto \theta_i^2\theta_j^2$). This connects naturally to the $>$ 2-qubit scaling results in Section 1.6.6.

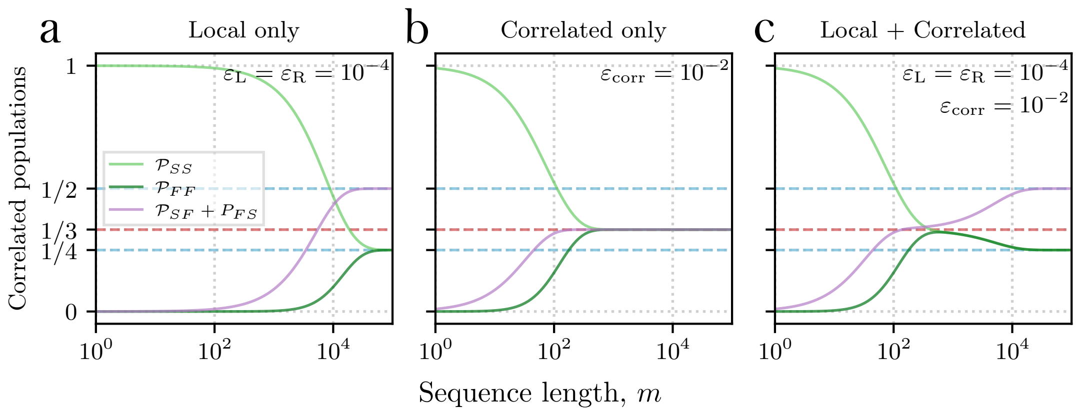

Figure 1.3: Numerical simulation of intermediate asymptotes measured in Z basis.

Correlated population decay

Using the block-diagonal Liouville structure of Eq. (**??**) and the scalar decay rates it implies, the probability of both qubits successfully returning to their initial state after m Cliffords, for a general error channel diagonal in the irrep basis of Table 1.1, is

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}\left(\alpha_L^m + \alpha_R^m\right) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \tag{1.29}$$

with corresponding expressions for $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$ following the same structure with sign changes on the α_L^m, α_R^m and $\alpha_{dt}^m, \alpha_{tr}^m$ terms respectively.

Combining the local depolarising channel (Eq. (**??**)) with the twirled common-noise channel (Section 1.3.3) gives, to first order, the sector decay rates

$$\alpha_L = 1 - \varepsilon_L - \varepsilon_{\mathrm{corr}}, \qquad \alpha_R = 1 - \varepsilon_R - \varepsilon_{\mathrm{corr}}, \tag{1.30}$$

$$\alpha_{tr} = 1 - \varepsilon_L - \varepsilon_R, \qquad \alpha_{dt} = 1 - \varepsilon_L - \varepsilon_R - 3\varepsilon_{\mathrm{corr}}, \tag{1.31}$$

where $\varepsilon_{\mathrm{corr}}$ is defined by Eq. (**??**). Because the symmetric trace sector is exactly invariant under common rotation noise ($\alpha_{tr} = 1$ when $\varepsilon_L = \varepsilon_R = 0$), Eq. (1.29) predicts an intermediate asymptote, $\mathcal{P}_{SS}|_{m\to\infty} = 1/3$, distinct from the fully depolarised asymptote $\mathcal{P}_{SS}|_{m\to\infty} = 1/4$ reached once any finite local error rate is present. This intermediate asymptote is the central experimental signature of correlated noise exploited throughout this chapter, and corresponds to the two-qubit state

$$\rho_{sym} = \frac{1}{3}\Big(|00\rangle\langle 00| + |11\rangle\langle 11| + |\Psi^+\rangle\langle\Psi^+|\Big), \tag{1.32}$$

where $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ is a Bell state. The presence of this Bell-state component reflects residual coherence induced between the two qubits by the shared rotation error, and motivates the independent, coherence-sensitive probe developed in Section 1.4.1.

Insert the full derivation connecting Eq. (1.29) to the vectorised-state inner product $\mathcal{P}_{L,R} = \langle\langle L, R|\Lambda_T^m|00\rangle\rangle$ and the explicit transfer matrices, rather than quoting the result (this is worked in the manuscript SM and can be reproduced here in full for thesis completeness, including the $\mathcal{P}_{0,1}, \mathcal{P}_{1,0}, \mathcal{P}_{1,1}$ companion expressions).

Standard Clifford RB also reveals a state-preparation and measurement, “SPAM", error as an initial scaling factor of the exponential decay. The effects of SPAM are discussed in section 1.3.8. The analysis for ZZ-type entangling errors, although not shown experimentally in this work, is also discussed below.

1.3.6 ZZ-type noise

Discuss ZZ-type errors, non-unitals, generalised depolarising channels, Stochastic vs coherent errors, etc. The simplifying assumption of single qubit errors results in a diagonal matrix in the Schur representation for any rotation angle. Here we describe the possibility to extend this analysis to include ZZ-type errors, described by the Hamiltonian,

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z, \tag{1.33}$$

where J_{ZZ} is undesired coupling strength.

Generally, H_{ZZ} will couple the subrepresentations of the 3-irrep, given in table 1.1. We note that the resulting block diagonal form is inconvenient to express as a general multi-exponential decay parametrised by small single qubit rotations and two-qubit couplings, J_{ZZ}, however, the resulting block diagonal Liouville form may be used directly with experimental data to fit these parameters of interest. We note that a convenient form is available for the stochastic variant of the ZZ-type error, given by,

$$\mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ}(Z \otimes Z)\rho(Z \otimes Z), \tag{1.34}$$

where p_{ZZ} is the probability of a correlated phase flip error on both qubits. In the irrep basis, the following sector decays are found for the stochastic ZZ error,

$$\begin{aligned} &\alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} = 1 - 4p_{ZZ}/3, \\ &\alpha_{tr} = \alpha_{dt} = 1. \end{aligned} \tag{1.35}$$

Comparing to correlated population decays given in Eq. 1.25, it can be seen that an intermediate asymptote of $\mathcal{P}_{SS}|_{m\to\infty} = 0.5$ is expected. This highlights that twirling over $Cl_1^{\otimes 2}$ both simplifies a generic error map to be measurable via RB, and, retains noise-dependent dynamics that can be used to discern spatially uncorrelated and correlated noise, as well as give evidence of the underlying error mechanism. Including other error mechanisms that lead to spatial correlations necessitates including more parameters to be extracted experimentally. For some parameter regimes, it can require many samples to accurately fit multi-exponential decays, however, we note that techniques utilised in *character RB* [18] may be implemented to k-copy RB: initial and final state choices are made to isolate individual sectors, thus only requiring the fitting of single exponential decays. This approach is analogous to the parity contrast measurements experimentally shown here to confirm the occupation of ρ_{sym} at the intermediate asymptote in Fig. **??**. A short discussion is presented in the following section, however we leave detailed exploration of this approach applied to ZZ-type errors to future work.

1.3.7 Correlated non-unital noise

1.3.8 State preparation errors

Similar to previous work on subspace leakage benchmarking [17], state preparation errors can lead to non-trivial modifications of the correlated population decay curves. First, we consider independent state preparation on two qubits. We model independent errors as probabilistic bit flips. To first order in error, on a single qubit we have,

$$\rho_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)}) \left|0\right\rangle \left\langle 0\right| + \varepsilon_{sp}^{(i)} \left|1\right\rangle \left\langle 1\right|, \tag{1.36}$$

where the superscript refers to which qubit, L, or R, the preparation error is relevant to. This error may be modelled by $\mathcal{E}_{sp}^{(i)}$, a probabilistic X channel applied to an arbitrary state,

$$\mathcal{E}_{sp}^{(i)}(\rho) = \left(1 - \varepsilon_{sp}^{(i)}\right) \rho + \varepsilon_{sp}^{(i)} X \rho X. \tag{1.37}$$

Transforming to the Liouville form, denoted by $\Lambda_{sp}^{(i)}$, this error is simply incorporated into the population dynamics given by Eq. 1.24, by acting on the initial state,

$$\mathcal{P}_{LR} = \langle\langle L, R | \Lambda_T^m \Lambda_{sp}^{(L)} \Lambda_{sp}^{(R)} | 00 \rangle\rangle. \tag{1.38}$$

In this picture, the state preparation errors scale only the overlap of the initial state with the sub-representations. The overlaps are multiplied by,

$$A_{sp}^{(i)} = (1 - 2\varepsilon_{sp}^{(i)}), \tag{1.39}$$

if the qubit is involved non-trivially in the subspace projector, i.e. for the left and right qubit sectors $A_{sp} = A_{sp}^{(i)}$. Relevant to this work, the correlated survival populations with state preparation errors included are,

$$\begin{aligned} \mathcal{P}_{SS} = \frac{1}{4} &+ \frac{A_{sp}^{(L)}}{4} \alpha_L^m + \frac{A_{sp}^{(R)}}{4} \alpha_R^m \\ &+ \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{6} \alpha_{dt}^m + \frac{A_{sp}^{(L)} A_{sp}^{(R)}}{12} \alpha_{tr}^m. \end{aligned} \tag{1.40}$$

Similar forms exist for populations, $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$. Significantly, this state preparation error does not add projection on the antisymmetric or symmetric off-diagonal sectors, which would introduce another exponential component into the observed decay curves.

Correlated state preparation errors may be considered by adapting Eq. 1.37 to a two-qubit stochastic error channel, such as correlated spin-flips given by

$$\mathcal{E}_{sp}^{corr}(\rho) = (1 - \varepsilon_{sp}^{corr})\rho + \varepsilon_{sp}^{corr}(X \otimes X)\rho(X \otimes X). \tag{1.41}$$

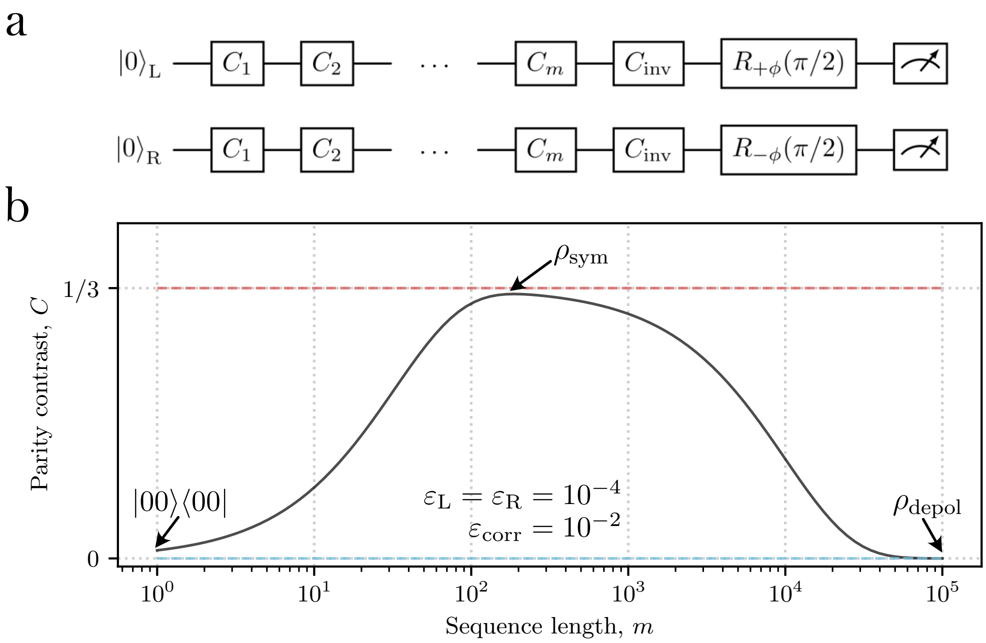

Figure 1.4: **circuit diagram for parity measurement. Could there also be a nice schematic of the Bloch sphere picture of the differential rotation?**

The subspaces involving both qubits non-trivially are now unaffected by the state preparation error, shown explicitly by,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{A_{sp}^{corr}}{4}\alpha_L^m + \frac{A_{sp}^{corr}}{4}\alpha_R^m + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \tag{1.42}$$

where $A_{sp}^{corr} = 1 - 2\varepsilon_{sp}^{corr}$. These correlations thus appear similar to *common noise* errors in the invariance of α_{tr} to this channel. Therefore correlated state prep errors can lead to a systematic bias in quantification of correlated error per Clifford if not accounted for in the fit by an initial state preparation error offset.

1.4 Isolating sectors

Liken to character RB, or partial tomography type test.

1.4.1 Parity contrast: an independent probe of coherence

Population decays alone indicate an intermediate asymptote; a parity-based measurement independently verifies that this asymptote corresponds specifically to occupation of the symmetric trace sector, by directly probing the residual coherence

$|01\rangle\langle 10|$ built up between the two qubits. Differential rotations of the form $\hat{R}_{+\phi}(\pi/2)\otimes\hat{R}_{-\phi}(\pi/2)$, with rotation axis $\hat{\sigma}_\phi$ in the XY-plane, are applied prior to measurement, and the contrast of the resulting fringe (as ϕ is varied) is extracted. This contrast has the predicted dependence

$$\mathcal{C}=\frac{1}{3}\left(\alpha_{tr}^m-\alpha_{dt}^m\right), \tag{1.43}$$

with a maximum value $\mathcal{C}_{\max}=1/3$, reached in the same regime in which the population asymptote of Eq. (1.29) sits at 1/3. Equations (1.29) and (1.43) may be combined to isolate the individual sector decay rates α_{tr} and α_{dt} as single exponentials, rather than fitting the multi-exponential population decay directly.

Fig. **??**(d) in the main text shows the contrast of parity measurements after a differential $\frac{\pi}{2}-$pulse is applied to the two qubits. This measurement directly probes coherences $|01\rangle\langle 10|$ and $|10\rangle\langle 01|$. The analytic model of the parity contrast (black dashed line), can be simply calculated by extending Eq. 1.24, to include a final differential rotation, which we refer to as the parity pulse, in transfer matrix form, Λ_Π,

$$\begin{aligned}\Pi(\phi)&=\langle\langle\pi|\Lambda_\Pi(\phi)\Lambda_T^m|00\rangle\rangle,\\&=\mathcal{C}\cos(2\phi),\end{aligned} \tag{1.44}$$

where Π is the parity signal, $|\pi\rangle\rangle=|00\rangle\rangle+|11\rangle\rangle-|01\rangle\rangle-|10\rangle\rangle$, ϕ describes the rotation axis of the differential pulse, and $\mathcal{C}$ is the parity contrast. The differential parity pulses are parallel single qubit rotations $\hat{R}^{(L)}_{+\phi}(\pi/2)$ applied to the left qubit, and $\hat{R}^{(R)}_{-\phi}(\pi/2)$ to the right qubit, with rotation axis $\sigma_\phi=\cos(\phi)\sigma_x+\sin(\phi)\sigma_y$. The parity contrast, which measures the overlap between the final mixed state after RB, and the pure Bell state $|\Psi^+\rangle$, (this is a typical reduced tomography method, see Ref. [23]), has the form,

$$\mathcal{C}=\frac{1}{3}(\alpha_{tr}^m-\alpha_{dt}^m) \tag{1.45}$$

with α_{tr} and α_{dt} given in Eq. 1.23 in terms of $\{\varepsilon_L,\varepsilon_R,\varepsilon_{corr}\}$. The maximum contrast seen in Fig. **??**(b), $\mathcal{C}_{\max}=1/3$ after noise incident on the initial state $|00\rangle$ can be understood by directly inspecting the space of possible pure two qubit states reached through common only rotations. The general pure state after common rotations is the separable identical two-qubit states,

$$|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle \tag{1.46}$$

with $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$ for a single qubit general state. Depolarising into the symmetric separable two-qubit states is equivalent to averaging over the full Bloch sphere,

$$\begin{aligned}\bar{\rho}_{\text{sym}} &= \frac{1}{4\pi}\int_0^{\pi}\int_0^{2\pi}\left(\sin(\theta)\,|\Psi\rangle\langle\Psi|\right)\mathrm{d}\phi\mathrm{d}\theta \\ &= \frac{1}{3}|00\rangle\langle 00| + \frac{1}{3}|11\rangle\langle 11| + \frac{1}{3}|\Psi^+\rangle\langle\Psi^+|\end{aligned} \tag{1.47}$$

where the third term is the Bell state $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$, with the expected $\frac{1}{3}$ population.

As noted in the main text, the advantage of considering the residual coherence of the decayed state is apparent when combined with the computational state measurements made in k-copy RB. Namely, if we define the computational parity, Π_Z, as

$$\begin{aligned}\Pi_Z &= \mathcal{P}_{SS} + \mathcal{P}_{FF} - \mathcal{P}_{SF} - \mathcal{P}_{FS} \\ &= \frac{1}{3}(2\alpha_{dt}^m + \alpha_{tr}^m),\end{aligned} \tag{1.48}$$

then in combination with $\mathcal{C}$, we may isolate single exponential decays originating from the independent sectors named "symmetric trace", and "diagonal traceless". Specifically the combinations

$$\begin{aligned}\Pi_Z - C &= \alpha_{dt}^m, \\ \Pi_Z + 2C &= \alpha_{tr}^m,\end{aligned} \tag{1.49}$$

result in isolating single decay constants.

Reproduce the full parity-contrast derivation from the manuscript SM (App. “Parity analysis”): the transfer-matrix expression $\Pi(\phi) = \langle\langle\pi|\Lambda_\Pi(\phi)\Lambda_T^m|00\rangle\rangle = \mathcal{C}\cos(2\phi)$, the Bloch-sphere-averaging argument giving $\mathcal{C}_{\max} = 1/3$, and the isolation identities $\Pi_Z - C = \alpha_{dt}^m$, $\Pi_Z + 2C = \alpha_{tr}^m$ using the computational parity $\Pi_Z = \mathcal{P}_{0,0} + \mathcal{P}_{1,1} - \mathcal{P}_{0,1} - \mathcal{P}_{1,0}$. This is a good candidate for a fully worked thesis subsection given its role in confirming the central experimental claim.

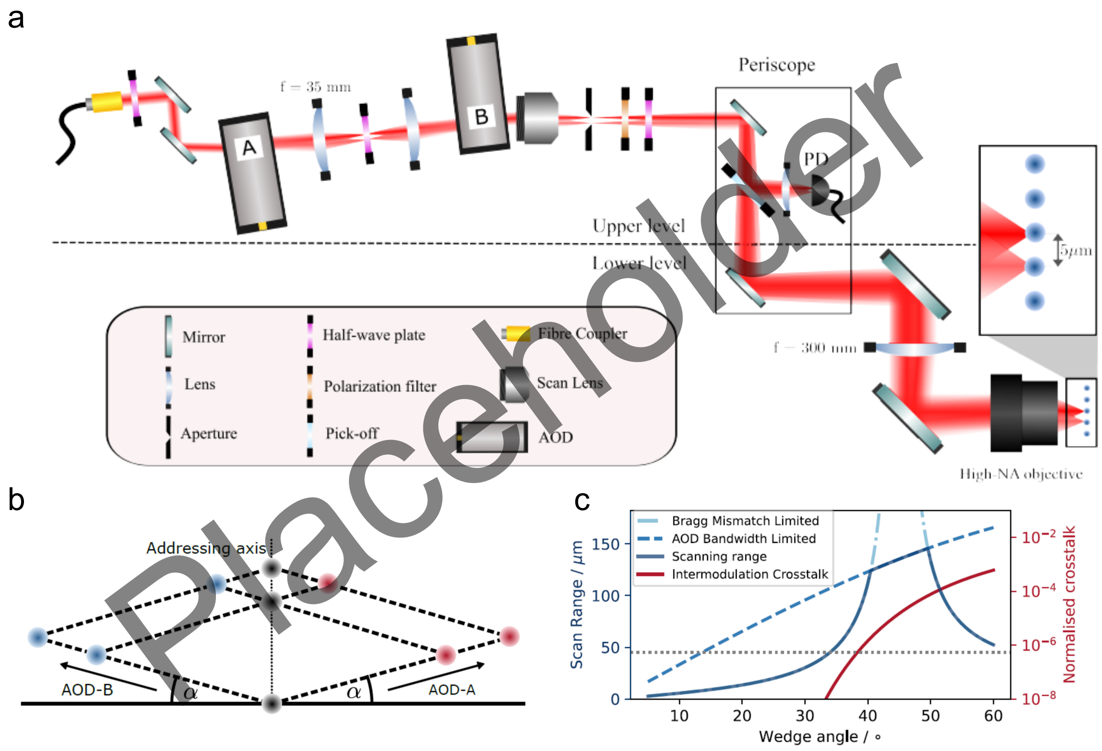

Figure 1.5: **Block diagonal Liouville structure of the two-copy Clifford twirl, showing the sub-representations and their associated decay rates.**

1.4.2 Singlet initial state

Beyond parity measurements, the sector decomposition of Table 1.1 suggests that preparing initial states with overlap on only a subset of sub-representations can isolate individual decay constants directly, without needing to fit multi-exponential curves. For example, preparing the singlet state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ gives a density matrix with overlap only on the identity and symmetric-trace sub-representations, so that the correlated populations decay as a single exponential in α_{tr} alone, insensitive to $\varepsilon_{\mathrm{corr}}$.

In the main text, we considered the initial prepared state $|00\rangle$, however, if we extend to the preparation of arbitrary initial states, we can isolate subsets of the scalar parameters $\vec{\alpha}$. This can be especially useful to simplify the decays curves from multiple exponentials to instead a single exponential decay. This can dramatically reduce the number of samples required for accurate fitting.

For example, consider the singlet state,

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \tag{1.50}$$

with density matrix in the Pauli basis,

$$\rho_s = \frac{1}{4}(II - XX - YY - ZZ). \tag{1.51}$$

By a change of basis to the irrep vectors given in table 1.1, it is clear to see that ρ_s has initial overlap with only the identity and "symmetric trace" sub-representations. Therefore, under the noise model considered in section **??**, we can immediately see that correlated populations decay with a single exponential,

$$\begin{aligned} \mathcal{P}_{SS} = \mathcal{P}_{FF} &= \frac{1}{4} - \frac{1}{4}\left(1 - 2\left(\varepsilon_L + \varepsilon_R\right)\right)^m, \\ \mathcal{P}_{SF} = \mathcal{P}_{FS} &= \frac{1}{4} + \frac{1}{4}\left(1 - 2\left(\varepsilon_L + \varepsilon_R\right)\right)^m, \end{aligned} \tag{1.52}$$

which corresponds to the scalar parameter α_{tr}.

This subsection is included as a brief pointer to a technique discussed only in outline in the manuscript SM (by analogy with *character RB* [18]). Expand into a full worked treatment if this state-engineering approach is pursued experimentally; otherwise keep as a short "future extensions" note and cross-reference from the Outlook (Section 1.7).

1.5 Scaling to $k > 2$ qubits

The main text considers pairwise measurements to detect and quantify spatial correlations between qubits. This technique is inherently efficient and scalable to many qubits as single-shot survival populations of a k-qubit register can be collected simultaneously, with pairwise comparisons being made in a classical post-processing step. However, this limits the technique to detecting and quantifying only 2-body correlations. We define the weight of a crosstalk error as the number of qubits, w, comprising a subsystem that cannot be decomposed into a convex combination of single qubit channels, when the full system not under the action

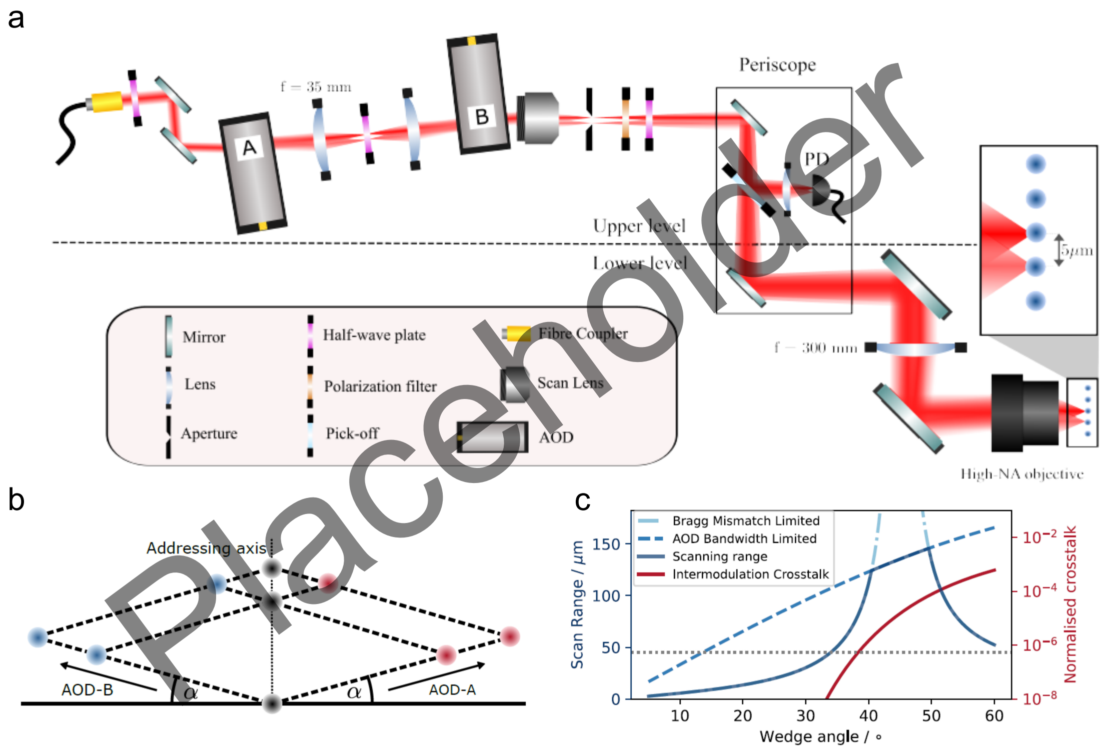

Figure 1.6: Asymptotic population decay for $k > 2$ qubits via numerical simulation.

of the crosstalk channel, is a convex combination of single qubit channels. The weight of physical error channels highly depends on processor type and topology, however, it is suggested in Ref. [5], that many physically relevant spatial correlation errors are low-weight. As quantum computing platforms continue to scale, it will soon become apparent to the engineer what error channels dominate in physical implementations.

We expect the symmetry resolved dynamics revealed by pairwise k-copy RB to generalise to higher weight errors, however, without prior work on the subrepresentation structure of the single-qubit Clifford group applied diagonally to > 2 qubits, extending the analysis for k-copy RB is challenging. However, for the *common noise* error model discussed throughout this work, a simple relation can be found for the asymptotic states. Namely, under *common noise* the state is driven to the maximally mixed state on the symmetric subspace, as this is the SU(2)-invariant

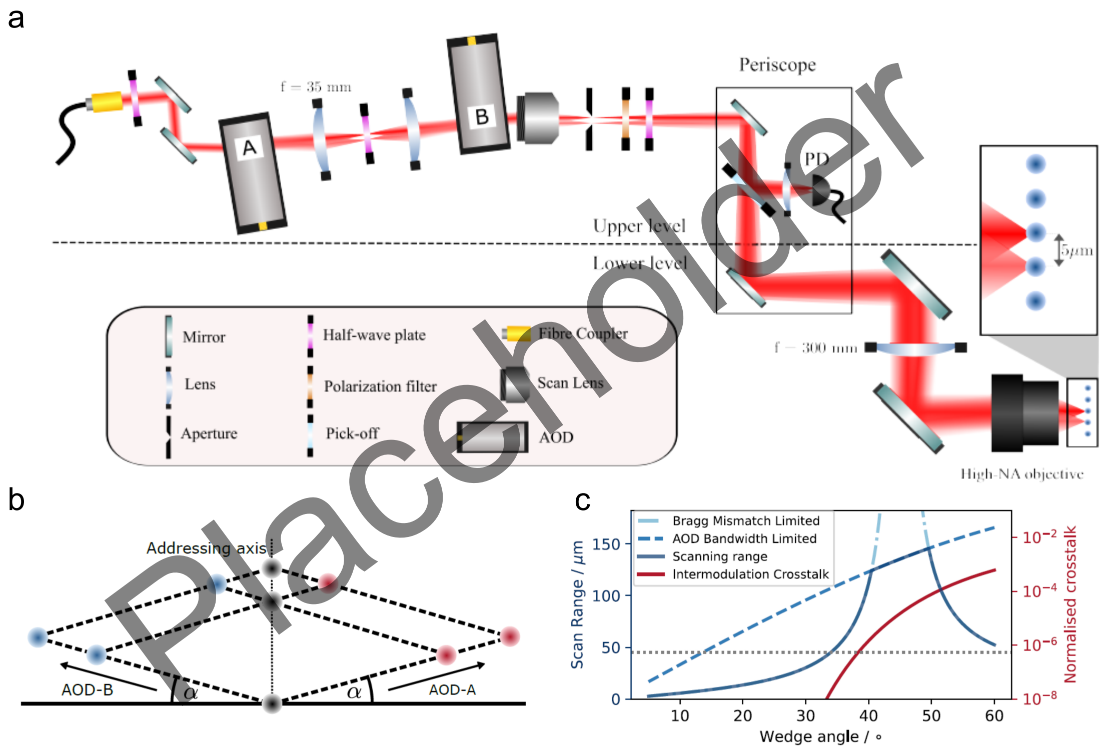

Figure 1.7: Experimental data and analytic fit for 3-qubits subject to correlated random-unitary single-qubit noise injected via random phase jitter ($\sigma_\phi = 0.06$ turns).

state on the symmetric irrep. This state is given by,

$$\rho_{\text{sym}} = \frac{1}{k+1}\sum_{l=0}^{k} |D_l^k\rangle\langle D_l^k|, \tag{1.53}$$

where $|D_l^k\rangle$ is the Dicke state for l excitations (here successful recoveries of target state), out of k-qubits. From this expression we see that an asymptote of $\mathcal{P}_{SS\ldots S}|_{m\to\infty} = 1/(k+1)$ is expected under *common noise*, while the maximally mixed state under local depolarisation leads to the familiar asymptote of $\mathcal{P}_{SS\ldots S}|_{m\to\infty} = 1/(2^k)$. This result suggests that the presence of *common noise* affecting many qubits may be simply observed, however, without the explicit representation structure, it cannot be easily quantified.

For three-qubits, we leverage symbolic computation to diagonalise the resulting PTM representation of *common noise* and recover analytic decays that quantify spatial correlations without deriving the representation structure. As an example, we show in Fig. 1.13 experimental data for three-qubits subject to *common noise*, with an analytic fit parametrised by local depolarising channels for each qubit, and a single 3-body *common noise* channel (note that this simple model excludes 2-body correlations). Here, $\mathcal{P}_{[SSF]}$ and $\mathcal{P}_{[SFF]}$ represent the sum of the permutations of single- and two-qubit failures respectively. Explicitly, $\mathcal{P}_{[SSF]} = \mathcal{P}_{SSF} + \mathcal{P}_{SFS} + \mathcal{P}_{FSS}$.

$$\alpha_1 = (1 - \varepsilon_1)^m, \tag{1.54}$$

$$\alpha_2 = (1 - \varepsilon_2)^m, \tag{1.55}$$

$$\alpha_3 = (1 - \varepsilon_3)^m, \tag{1.56}$$

$$\beta = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3, \tag{1.57}$$

$$\alpha_{\text{corr1}} = (1 - \varepsilon_{\text{corr}})^m, \tag{1.58}$$

$$\alpha_{\text{corr2}} = (1 - 3\varepsilon_{\text{corr}})^m, \tag{1.59}$$

$$\alpha_{\text{corr3}} = (1 - 6\varepsilon_{\text{corr}})^m, \tag{1.60}$$

$$M_1 = \beta, \tag{1.61}$$

$$M_2 = 3(\alpha_1 + \alpha_2 + \alpha_3)\alpha_{\text{corr1}}, \tag{1.62}$$

$$M_3 = 2\beta\alpha_{\text{corr2}}, \tag{1.63}$$

$$M_4 = \frac{6}{5}\alpha_1\alpha_2\alpha_3\alpha_{\text{corr3}}, \tag{1.64}$$

$$M_5 = \frac{9}{5}\alpha_1\alpha_2\alpha_3\alpha_{\text{corr1}}, \tag{1.65}$$

$$\mathcal{P}_{SSS} = \frac{1}{8} + \frac{1}{24}(M_1 + M_2 + M_3 + M_4 + M_5), \tag{1.66}$$

$$\mathcal{P}_{FFF} = \frac{1}{8} + \frac{1}{24}(M_1 - M_2 + M_3 - M_4 - M_5), \tag{1.67}$$

$$\mathcal{P}_{[SSF]} = \frac{3}{8} + \frac{1}{24}(-M_1 + M_2 - M_3 - 3M_4 - 3M_5), \tag{1.68}$$

$$\mathcal{P}_{[SFF]} = \frac{3}{8} + \frac{1}{24}(-M_1 - M_2 - M_3 + 3M_4 + 3M_5), \tag{1.69}$$

Limitation here is not including weight-2 errors.

1.6 Experimental Results

The results presented here constitute the first project-data output from the newly commissioned *FastGates* apparatus. This work was both necessary for confirming the utility of the *k-copy* RB protocol, as well as pushing the automation of the apparatus for robust operation, useful for all future experimental projects.

1.6.1 Physical implementation

We demonstrate *k-copy* RB using up to eight trapped $^{40}\mathrm{Ca}^+$ ions separated by an average spacing of $\sim$ 5 µm. The qubit is encoded in the optical transition $|4S_{1/2}, m_j = -\frac{1}{2}\rangle \leftrightarrow |3D_{5/2}, m_j = -\frac{5}{2}\rangle$ in a quantising field of $\sim$ 5 G. Single-qubit rotations are driven by 729-nm light, with details given in section **??**. The two-qubit coherence and Stark shift measurements use the individual addressing system, detailed in section **??**, while all other measurements use the global elliptical beam, detailed in section **??**.

The single-qubit Clifford gate set is decomposed into native $\pi/2$ single-qubit rotations of the form $\hat{R}_x(\pi/2)$ and $\hat{R}_z(\pi/2)$: $\hat{\sigma}_x$ rotations are performed natively on resonance with the qubit transition, with relative phase set by modulating the 729-nm beam via an acousto-optic modulator (AOM), while $\hat{\sigma}_z$ rotations are performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations. Two decomposition strategies are used, crucially with varying average numbers of physical $\hat{\sigma}_x$ $\pi/2$-gates per Clifford. The first uses an average of $\kappa = 2.17$ $\pi/2$-gates, while the second uses $\kappa = 1.167$ gates.

We begin by reproducing the multi-exponential decays seen in simulation, figure 1.3, followed by parity measurements to observe the coherence of the intermediate symmetric state described in section 1.4.1. We then present data with known injected errors through phase and detuning offsets to show the ability for *k-copy* RB to discern correlated and uncorrelated errors for a range of magnitudes. We finish this section with evidence for scaling the protocol to more than two qubits. We show both data for three-qubit decay profiles, and pairwise analysis of an eight-ion chain with dominant correlated errors due to global drive field inhomogeneity.

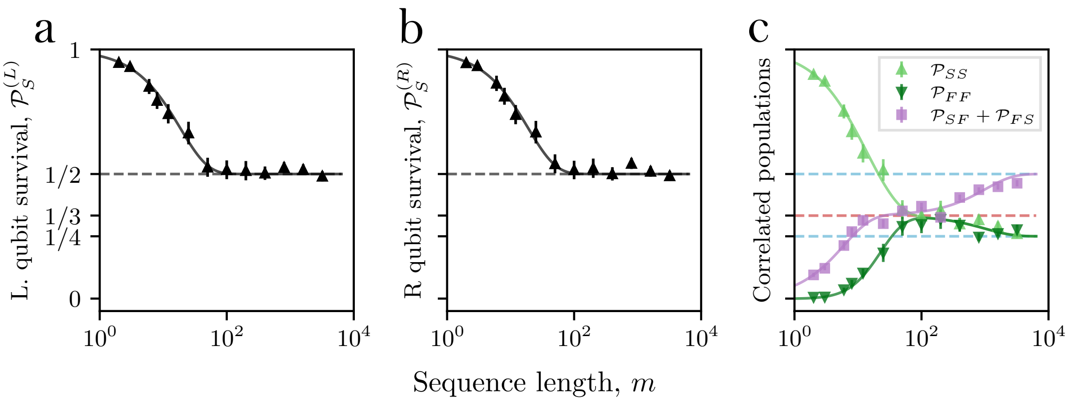

Figure 1.8: Correlated two-qubit probabilities after k-copy RB. Panels (a,b) show the single-qubit success probabilities, which decay with a single exponential (solid line) toward $\mathcal{P}_S|_{m\to\infty} = 1/2$. Panel (c) shows the correlated two-qubit populations, which exhibit an intermediate $\mathcal{P}_{SS} = 1/3$ asymptote before relaxing toward $\mathcal{P}_{SS}|_{m\to\infty} = 1/4$.

1.6.2 Z-basis measurements

The k-copy RB protocol, with circuit diagram shown in figure 1.1, is executed on two qubits using a global field. To ensure a visually clear intermediate asymptote is seen, we set a fixed 10 kHz frequency detuning of the 729-nm laser compared to the bare qubit. As single qubits gates are calibrated for on-resonance pulses, this detuning will be a large *common noise* source. The Clifford sequence length is log-sampled up to $m = 3200$ total gates to capture both the intermediate asymptote due to this common unitary error, and the final decay to the fully depolarised state. The recovered multi-exponential decay from the correlated success and failure rates of the two qubits is shown in figure 1.8(c). Each solid point is inferred through $r = 50$ randomisations and $s = 50$ shots per randomisation. For each random sequence j, the population $\mathcal{P}_{SS,j}$ is estimated from the s binary measurement outcomes, $P_{SS,j} = \frac{1}{s}\sum_{i=1}^{s} X_{ij}$, where $X_{ij} = 1$ if the ith shot is measured in state SS, and $X_{ij} = 0$ otherwise. The recovered population is then the mean over the r random sequences, $\mathcal{P}_{SS} = \frac{1}{r}\sum_{j=1}^{r} \mathcal{P}_{SS,j}$. Error bars show the standard error of the mean across random sequences,

$$\mathrm{SE}(\mathcal{P}_{SS}) = \sqrt{\frac{1}{r^2}\sum_{j=1}^{r} (\mathcal{P}_{SS,j} - \mathcal{P}_{SS})^2}. \tag{1.70}$$

The correlated populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|00\rangle$ approaching ρ_{sym}, equation 1.32, with $\mathcal{P}_{SS} \to \frac{1}{3}$ asymptote highlighted by the red dashed line. The dashed blue line shows the fully depolarised $\mathcal{P}_{SS} \to \frac{1}{4}$ asymptote, seen clearly by $m \approx 1000$ gates. The solid lines are found through modelling errors by a combination of local depolarising errors and *common noise*, given by equation 1.25 generalised to include independent state-preparation and measurement error (SPAM), ε_{sp} (see equation. 1.40 for example). The model is fit using the non-linear Levenberg–Marquardt method, yielding extracted parameters $\varepsilon_{sp} = 0(5) \times 10^{-3}$, $\varepsilon_{\text{corr}} = 2.6(2) \times 10^{-2}$, $\varepsilon_L = 5(6) \times 10^{-4}$, and $\varepsilon_R = 0(6) \times 10^{-4}$, confirming that correlated errors dominate with this global field detuning. [2]

From the same shot data record we recover the single-qubit populations required for standard RB, as described in section **??**. To match the naming of the relevant participating irreps, we label the two qubits L and R for left and right respectively. The standard RB single exponential decay curves are shown in (a) and (b), with extracted single-qubit error per Cliffords of $\varepsilon_c^{(L)} = 2.6(2) \times 10^{-2}$ and $\varepsilon_c^{(R)} = 2.6(2) \times 10^{-2}$. As expected, *k-copy* RB provides evidence for this standard error per Clifford to be decomposed to local and correlated components, given by equation 1.28.

1.6.3 Coherence measurements

A partial-tomography parity measurement, given by the circuit seen in figure 1.4(a), is used to independently confirm occupation of the symmetric sector, quantifying overlap of the final state with Bell state, $|\Psi^+\rangle$. As the final differential rotation pulse requires local addressing of the ions, the single addressing system is used. Each random sequence is repeated for varying final rotation angle, ϕ, to produce parity fringes, examples shown in (c) and (d) in figure 1.9 for $m = 25$ and $m = 1600$ Clifford gates respectively. The contrast is extracted by equation 1.45, and a plot against sequence length is shown in (b). The contrast asymptote $\mathcal{C} \to 1/3$ is observed for

[2] The large relative uncertainty of both local depolarising rates is due to the large covariance between ε_L and ε_R. The fitting may be constrained to the sum of local errors, to find $\varepsilon_L + \varepsilon_R = 5(1) \times 10^{-4}$. However, if precision measurements of this local error is of interest, either r, and s could be increased, or the local error contributions could be isolated as discussed in section 1.4.2.

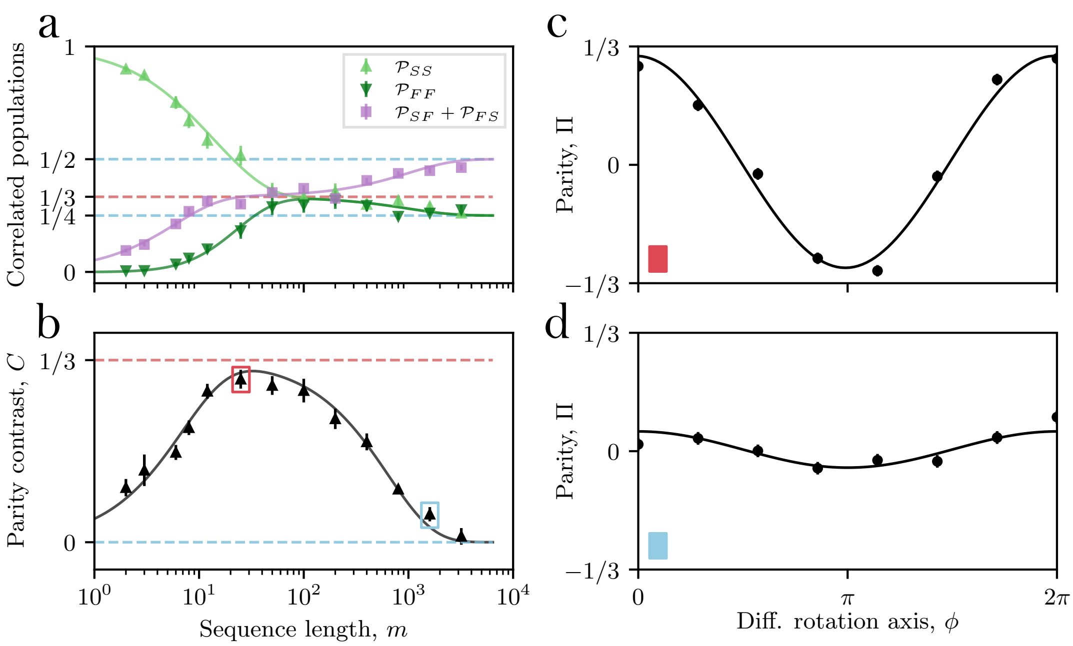

Figure 1.9: Parity contrast versus sequence length; (a) shows the standard Z measured correlated populations. (b) shows parity contrast — a signal proportional to occupation of the symmetric-trace sector. Solid lines are multi-exponential decays given by Eq. 1.29 and Eq. 1.43 for (a) and (b) respectively. Dashed lines indicate the expected asymptotes. (c) and (d) show example parity fringes for sequences of $m = 25$ and $m = 1600$ Clifford gates respectively.

$m \in [10, 100]$, corresponding to the residual coherence of ρ_{sym}, equation 1.47. The parity contrast data is fit (solid line) according to equation 1.43 to extract error per Cliffords of, $\varepsilon_{\text{corr}} = 2.1(2) \times 10^{-2}$ and $\varepsilon_L = \varepsilon_R = 3.9(3) \times 10^{-4}$.

1.6.4 Controlled noise injection

To verify the ability for *k-copy* RB to discern correlated and uncorrelated errors, we compare the magnitude of extracted errors to known injected errors. We limit the injected error method to those consisting of noise on common drive fields, and local depolarising type noise. Both of these noise models can be induced entirely through single qubit rotations, which are trivial to implement on our device.

Laser phase jitter

We first inject correlated errors via phase noise on a common drive field. This noise mechanism is prevalent in many platforms that use a common oscillator, such as laser, microwave, or radio-frequency (RF) fields to resonantly drive qubit transitions.

The injected error magnitude is inferred through considering the ideal and noisy unitary channel. A unitary qubit rotation is given by,

$$\hat{U} = e^{-i\frac{\theta}{2}\vec{n}\cdot\vec{\sigma}}, \tag{1.71}$$

where θ is the rotation angle, and $\vec{n}\cdot\vec{\sigma}$ is the rotation axis. We decompose the single-qubit Clifford gate set into a series of resonant $\theta = \pi/2$ qubit rotations around axes $\hat{\sigma}_x$ and $\hat{\sigma}_z$. Practically, phase noise on either the qubit frequency or drive field will introduce some deviation of this rotation axis. We shall consider the effect of zero-mean phase noise on $\hat{\sigma}_x$ rotations, leading to an erroneous rotations axis,

$$\hat{\sigma}_{\delta\phi} = \cos(\delta\phi)\hat{\sigma}_x + \sin(\delta\phi)\hat{\sigma}_y, \tag{1.72}$$

where $\delta\phi$ is the standard deviation of the noise in radians. We may find the expected error of a single Clifford gate using,

$$\varepsilon_{\delta\phi} = \kappa\left(1 - \mathcal{F}(V,U)\right), \tag{1.73}$$

where κ is the average number of $\pi/2$ rotations in a Clifford gate, and $\mathcal{F}(V,U)$ is the average gate fidelity. The fidelity of two unitaries averaged over the Haar measure is given in appendix equation A.3, reproduced here,

$$\mathcal{F}(V,U) = \frac{1}{d(d+1)}\left(|\mathrm{Tr}(U^\dagger V)|^2 + d\right). \tag{A.3}$$

Considering $\theta = \pi/2$ rotations, we find,

$$\hat{U} = e^{-i\frac{\pi}{4}\hat{\sigma}_x} = \frac{1}{\sqrt{2}}\left(\hat{I} - i\hat{\sigma}_x\right), \tag{1.74}$$

$$\hat{V} = e^{-i\frac{\pi}{4}\hat{\sigma}_{\delta\phi}} = \frac{1}{\sqrt{2}}\left(\hat{I} - i\hat{\sigma}_{\delta\phi}\right). \tag{1.75}$$

The fidelity, given that dimension $d = 2$ for a single qubit, is found to be,

$$\mathcal{F}(V,U) = \frac{1}{6}\left((1+\cos(\delta\phi))^2 + 2\right). \tag{1.76}$$

For small phase noise standard deviations, $\delta\phi \ll 1$, we find an expected error per Clifford of,

$$\varepsilon_{\delta\phi} = \frac{\kappa}{3}(\delta\phi)^2. \tag{1.77}$$

A known phase error may be conveniently injected through phase offsets sampled from a normal distribution with zero mean and standard deviation $= \delta\phi$ applied to the RF drive of a control AOM on the 729-nm global addressing path. The phase of the AOM RF is imprinted directly onto the 729-nm light leading to the expected drop in fidelity of the Clifford gates. In this experiment $\kappa = 2.17$ $\pi/2$ rotations. A full *k-copy* RB experiment is executed with logarithmic sampling up to $m = 512$ Clifford gates with $r = 50$ randomisations and $s = 50$ repeats. The resulting multi-exponential success and failure probability decay curves are fit to equation **??**, again generalised to include state-preparation errors. The correlated and local error components are found for a range of injected phase noise standard deviations. The resulting measured errors are shown in figure 1.10. The x-axis is the expected error injected $\varepsilon_{\delta\phi}$, while the y-axis is the extracted error magnitudes. The fit, shown by the solid red line, is a direct proportionality of injected error plus a baseline correlated error rate not due to phase injection,

$$\varepsilon_{\text{corr}} = \varepsilon_{\text{corr},0} + \varepsilon_{\delta\phi}. \tag{1.78}$$

The baseline correlated error is therefore the only parameter to fit. With $varepsilon_{\text{corr},0} = 1.71(4) \times 10^{-3}$, we find good agreement between the model and data. The local depolarising error on both qubits remains relatively constant, with the left qubit having a larger local error compared to the right qubit. We believe this asymmetry in local error is due to the global 729-nm beam not being centrally aligned to both qubits. The uncertainty in extracted error is found through Levenberg-Marquadt non-linear fit, it can be seen that the uncertainty in the errors increases with the total correlated error. **Compare this with analytic result when multi-exps are of different magnitudes.**

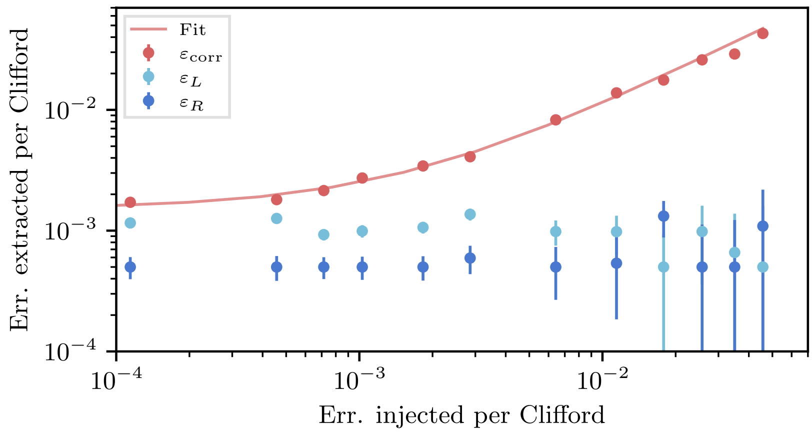

Figure 1.10: Injecting correlated noise between two qubits via common drive phase noise. Points correspond to fitted error per Clifford from full *k-copy* RB experiments. The solid red line is a linear fit between measured and injected error.

Laser detuning scan

k-copy RB can isolate errors arising from shared control fields, making it useful both for device characterisation and for optimising global control parameters. A fixed detuning of the global drive, under the action of *k-copy* Clifford twirling, is well modelled by the common random unitary error channel described in section 1.3.3. We demonstrate this by scanning the 729-nm laser frequency used for single-qubit rotations away from the bare qubit resonance. As with the previous section, we consider the fidelity of the error channel compared to the ideal $\pi/2$ $\hat{\sigma}_x$ single qubit rotation.

A coherent detuning error is written as,

$$V = \exp\left(-i\frac{\pi}{4}\left(\hat{\sigma}_x + \frac{\delta}{\Omega}\hat{\sigma}_z\right)\right), \tag{1.79}$$

where δ is the detuning from resonance, and Ω is the Rabi frequency. Assuming $\delta/\Omega \ll 1$, the fidelity is found to be,

$$\mathcal{F}(V,U) = 1 - \frac{1}{3}\left(\frac{\delta}{\Omega}\right)^2 + \mathcal{O}\left(\frac{\delta}{\Omega}\right)^4. \tag{1.80}$$

To convert this to an error contribution to a single qubit Clifford gate, we introduce the scalar κ_δ, to account for both the number of $\pi/2$ rotation in each Clifford gate,

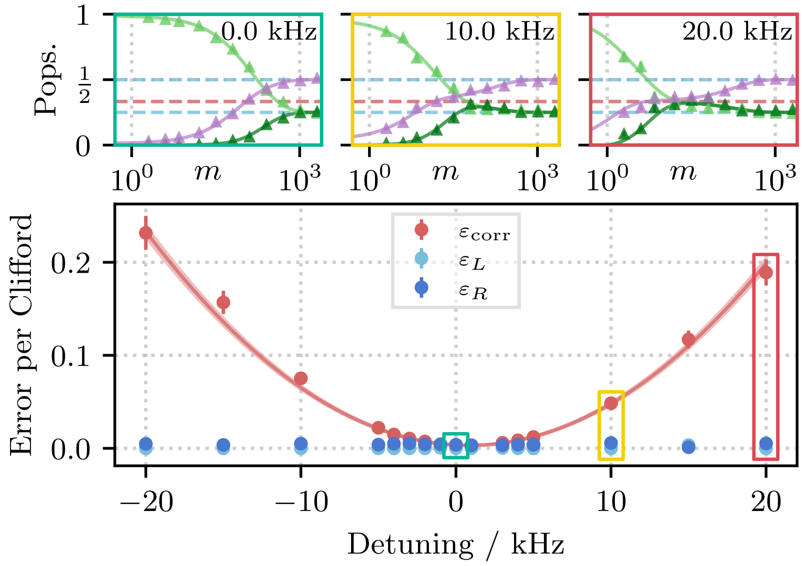

Figure 1.11: Correlated and single-qubit depolarising errors extracted as a function of shared-drive detuning. The correlated error (red points) rises quadratically with laser detuning, while the local depolarising errors (blue points) remain approximately constant. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. The red shaded region signifies the 68% confidence, generated through covariance-based parameter sampling.

but also the duration and delays between pulses where phase tracking between qubit and laser also drift. The error contribution is then given by

$$\varepsilon_\delta = \kappa_\delta(1 - \mathcal{F}(V, U)) = \frac{\kappa_\delta}{3}\left(\frac{\delta}{\Omega}\right)^2 + \mathcal{O}\left(\frac{\delta}{\Omega}\right)^4. \tag{1.81}$$

Each data point is obtained from a complete *k-copy* RB experiment with 11 sequence lengths up to $m = 1024$, using $r = 50$ random sequences and $s = 50$ shots per sequence, with correlated populations fit to equation 1.29 to extract $\{\varepsilon_{\mathrm{corr}}, \varepsilon_L, \varepsilon_R\}$ at each detuning. The upper panels of figure 1.11 show example datasets for the *k-copy* RB sequences, while the lower panel shows the extracted error per Cliffords as a function of detuning. The correlated error is observed to increase quadratically with detuning, while the local depolarising errors remain approximately constant, consistent with detuning of a shared drive producing *common noise*. To fully parameterise the injected error, we include a background correlated error not originating from detuning, ε_0, and a shift of the light-shifted qubit versus the bare

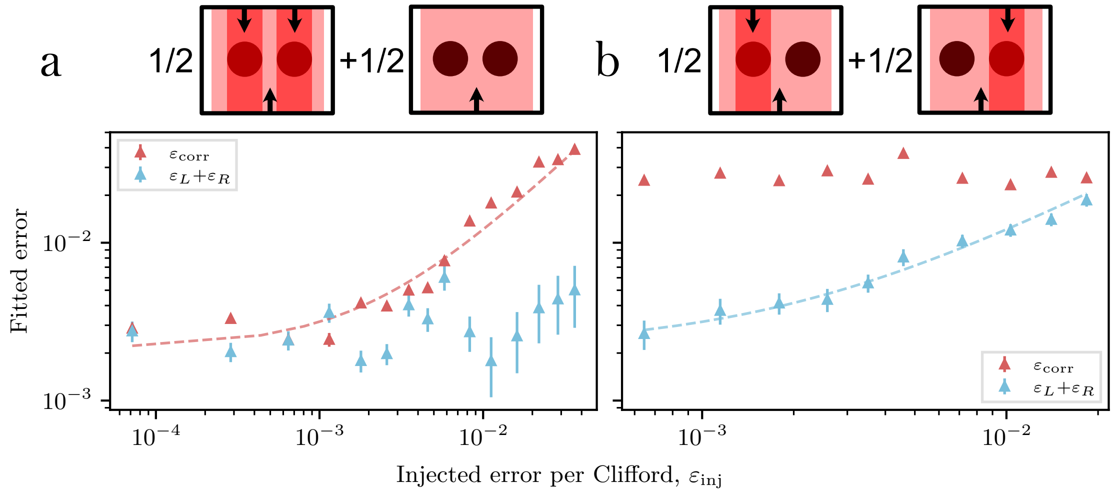

Figure 1.12: Correlated and uncorrelated errors are injected via a single-addressed Stark-shift beam, and quantified through fitting the correlated success populations. The four addressing configurations are depicted visually above the plot. The wide beam signifies the on-resonance global drive which executes the parallel single-qubit rotations, while the narrow beams represent the off-resonant Stark shift beams addressing single-qubits (black circles). In (a) the errors are correlated between qubits, while panel (b) shows the extracted errors after independent errors are injected. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations.

qubit frequency, δ_0. The solid red line is therefore fit to the model,

$$\varepsilon_{corr} = \varepsilon_0 + \frac{\kappa_\delta}{3}\left(\frac{\delta_L - \delta_0}{\Omega}\right)^2. \tag{1.82}$$

Here, $\Omega = 2\pi \times 100(1)\,\text{kHz}$ is the independently measured Rabi frequency. Fitting gives $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $\kappa_\delta = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)\text{kHz}$.

Single addressed stark shifts

Uncorrelated noise injection requires the introduction of noise with spatial inhomogeneity. This is implemented experimentally through an off-resonant single addressing beam inducing Stark-shifts between the on-resonant parallel single qubit rotations executing the *k-copy* twirling. The single addressing beam is detuned from the qubit resonance by -15 MHz, and the induced Stark shift with beam power, γ, is calibrated using a Ramsey sequence, here measured as

$\gamma = 8.69 \times 10^7 \mathrm{Hz\,W}^{-1}$**uncertainty?**. The injected error, $\varepsilon_{\mathrm{inj}}$, is inferred by,

$$\varepsilon_{\mathrm{inj}} = \frac{\kappa}{3}\phi^2, \tag{1.83}$$

where the phase error, given in radians is, $\phi = P\tau\gamma$, with P being the power in the single addressing beam, τ the duration of the Stark-shift pulse. In these experiments, the duration of the Stark-shift pulse is fixed to $\tau = 20$ µs, while the power of the beam is varied to supply a known error.

Two experimental sequences were run, the first, shown in figure 1.12(a), implements stochastic *common noise* through applying either two spots shifting both ions, or no Stark shift, with a $50:50$ chance after each Clifford gate. The second experiment, shown in (b), applies local errors through a single spot either addressing the left or right qubit, again with $50:50$ chance after each Clifford gate. These two arrangements are shown schematically in the upper panels. To ensure correlated errors dominate even with injected local errors, an additional constant correlated error offset of $\varepsilon_{corr,0} \approx 2 \times 10^{-2}$ is applied through random phase fluctuations, as described in section 1.6.4. This reduces the number of samples required to accurately estimate the correlated and uncorrelated error contributions, which is of practical importance in collecting data in reasonable time (for reference, this data was collected over ~ 3 hours). It can be seen that $\varepsilon_{\mathrm{corr}}$ increases only when injecting *common noise*, and remains constant when local errors are injected. Dashed lines are simple linear fits given by $\varepsilon_{\mathrm{fit}} = \varepsilon_0 + \varepsilon_{\mathrm{inj}}$, where $\varepsilon_{\mathrm{fit}}$ are the extracted errors, ε_0 is some residual error baseline. These straight line fits show good agreement between measured and injected error when accounting for some fit baseline error rates.

1.6.5 Three-qubit decay

As an example, we show in figure 1.13 experimental data for three-qubits subject to *common noise*, with an analytic fit parametrised by local depolarising channels for each qubit, and a single 3-body *common noise* channel (note that this simple model excludes 2-body correlations). Here, $\mathcal{P}_{[SSF]}$ and $\mathcal{P}_{[SFF]}$ represent the sum of the permutations of single- and two-qubit failures respectively. Explicitly, $\mathcal{P}_{[SSF]} = \mathcal{P}_{SSF} + \mathcal{P}_{SFS} + \mathcal{P}_{FSS}$.

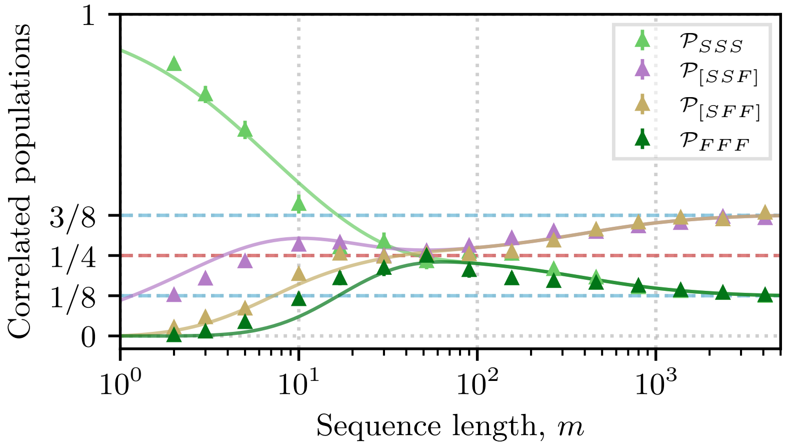

Figure 1.13: Experimental data and analytic fit for 3-qubits subject to correlated random-unitary single-qubit noise injected via random phase jitter ($\sigma_\phi = 0.06$ turns).

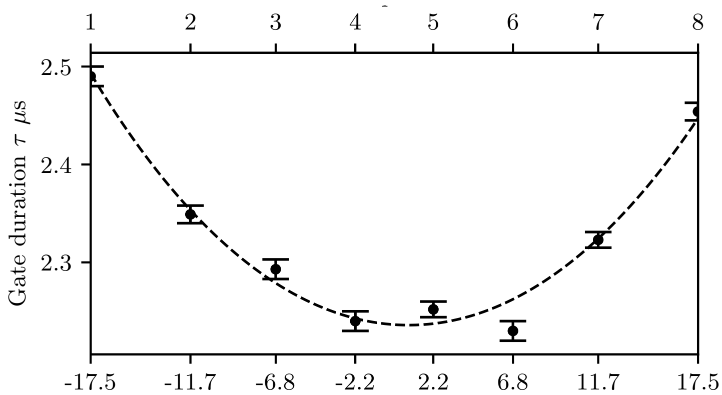

Figure 1.14: Variation of π-pulse gate duration along eight ion chain due to global beam Gaussian profile.

1.6.6 Scaling to a multi-ion register

Pairwise *k-copy* RB extends simply to > 2 qubits: correlated populations are fit to equation 1.29 independently for each qubit pair. For a register of k qubits, all $k(k-1)/2$ pairwise outcomes are extracted from the same shot record — no additional trials on the quantum register are required, though classical post-processing scales with the number of pair combinations. The protocol itself requires only synchronised application of identical single-qubit Clifford sequences and spatially resolved readout. This scaling is demonstrated on an eight-ion chain, with parallel single-qubit

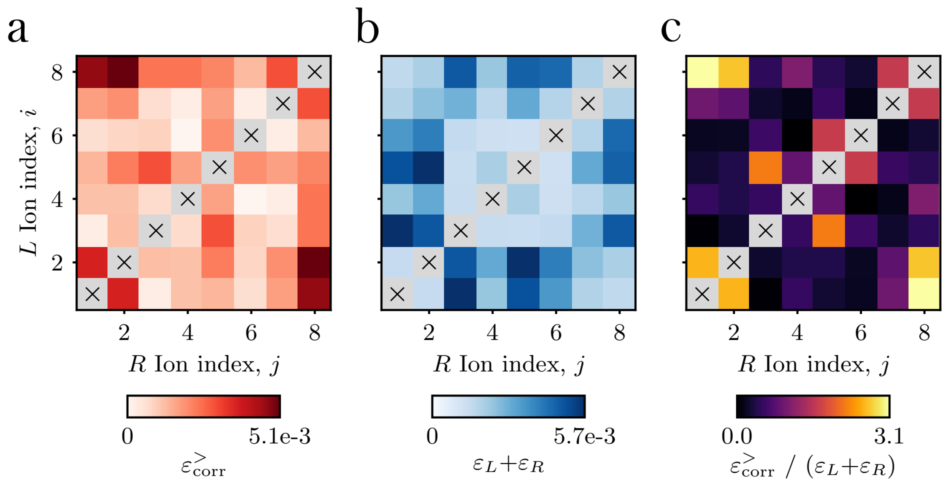

Figure 1.15: Correlated and local error per Clifford measured for all qubit pairs after applying k-copy RB to an 8 ion chain. Noise is dominated by position dependent duration of $R_x(\pi/2)$ gates consistent with inhomogeneous illumination by the global drive. Panel (a) displays the extracted correlated errors, $\varepsilon^{(i,j)}_{\mathrm{corr}}$ for each qubit pair, (i,j). Panel (b) displays the sum of both independent depolarising errors, $(\varepsilon_L+\varepsilon_R)^{(i,j)}$. Panel (c) displays the ratio of excess correlated noise (offset from phase noise injection), $(\varepsilon^{>}_{\mathrm{corr}})^{(i,j)} = \varepsilon^{(i,j)}_{\mathrm{corr}} - \varepsilon^{(0)}_{\mathrm{corr}}$ to the sum of local errors given in (b). This plot highlights the outer qubits are dominated by spatially correlated errors.

rotations driven by an elliptical 729-nm beam. We expect to observe structured errors due to the Gaussian intensity profile of the beam. To verify the expected gate duration dependence due to the finite beam, we run parallel gate duration calibration experiments (see section **??** for details). The gate duration, τ, as a function of qubit position in the beam is expected to follow,

$$\tau(r) \propto \frac{1}{\sqrt{I(r)}}, \tag{1.84}$$

where $I(r)$ is a Gaussian distribution, $I = a\exp\left(\frac{-(r-r_0)^2}{2c^2}\right)$. Figure 1.14 shows the measured π-pulse gate durations for each ion, with the fit given by the dashed line. From this, we find an expected major-axis waist radius of 27.7(9)µm aligned to the ion chain. The qubit spacing, on the lower axis of the calibration plot, is inferred through the axial confinement of 0.5 MHz using equilibrium positions of a linear chain given in [24]. At this confinement, the chain extent is 35 µm, giving an expected Rabi-frequency difference of 10.3(6)% between the outer and inner ions. We calibrate global single-qubit gates using the inner ion, $i = 4$, leading to an expected per Clifford error of (using fidelity definition from appendix A.5),

$$\varepsilon_{c,max} = \frac{2}{3}\kappa\left(\alpha\frac{\pi}{4}\right)^2 = 5.08(2) \times 10^{-3}, \tag{1.85}$$

where, here the Clifford decomposition used gives $\kappa = 1.167$, and the fractional Rabi frequency difference is given by the above calibration curve, $\alpha = 0.103(6)$.
As with the single qubit Stark-shift dataset, we ensure that correlated errors dominate to improve the fitting sensitivity. This is done through a fixed phase noise error corresponding to $\varepsilon_{corr,0} = 1.3 \times 10^{-2}$. In the following data, we subtract this baseline offset to highlight the excess correlated noise due to the beam structure, $\varepsilon^{>}_{corr} = \varepsilon_{corr} - \varepsilon_{corr,0}$.
Figure 1.15 shows the extracted pairwise correlations. The excess correlated error is shown in (a), with the largest values seen for the outer ion pairs, agreeing with the expected maximum error of $\varepsilon^{>}_{corr,max} = \varepsilon_{c,max}$. Panel (b) shows the sum of the local depolarisation error per Cliffords on both qubits, with the largest values now corresponding to pairs consisting of one central ion, and one outer ion. We plot the ratio of excess correlated error and depolarisation errors in panel (c) to further highlight the error distribution due to the beam shape.

1.7 Conclusions and outlook

k-copy RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. Because of the symmetry enforced by twirling over the diagonal Clifford representation $Cl^{\otimes k}$ (Section **??**), structured errors drive the decay of correlated populations into invariant subspaces in a way that is directly observable without full process reconstruction. Given a specified error model, these correlated decays can be used to directly quantify the magnitude of the individual error channels contributing to the total error per Clifford.

This method has been proposed and experimentally demonstrated, and its application extended to an eight-qubit register through pairwise comparisons (Section 1.6.6). This approach permits highly efficient parallel data collection:

the time required on the device is independent of qubit number, and the protocol requires exclusively local operations followed by classical post-processing of correlated survival rates.

Expand this Outlook section substantially for the thesis, drawing together the forward-looking threads seeded earlier in this chapter and in Chapter ??:

- **Extension to genuine ≥ 3-body correlated error terms, beyond the pairwise diagnostic of Section 1.6.6.**
- **Experimental characterisation of ZZ-type entangling errors (Section ??), not demonstrated here.**
- **Sample-cost / Fisher-information analysis for multi-exponential fitting — flagged as an open numerics question in the manuscript SM and not yet resolved.**
- **Non-Markovian (temporally correlated) noise, via repeated identical Clifford RB sequences applied at different times on a single qubit, exploiting the same correlated-outcome analysis used here for spatial correlations.**
- **Character-RB-style initial-state engineering (Section ??) to isolate single exponential decays and reduce sample cost.**

Appendices

A
Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Gate Fidelity

We derive the fidelity of a noisy gate implementation with respect to the ideal unitary.
A quantum channel,

$$\mathcal{E} : \rho \mapsto \rho, \tag{A.1}$$

which executes ideal unitary, U, with some error, has an average operation fidelity given by,

$$F(\mathcal{E}, U) = \int d\psi \, \langle\psi| \, U^\dagger \mathcal{E}(|\psi\rangle \langle\psi|) U \, |\psi\rangle \,, \tag{A.2}$$

where the integral is taken over the uniform Haar measure on the space of pure states. We shall see that the language used for this analysis of single qubit gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V,U) = \frac{1}{d(d+1)}\left(|\mathrm{Tr}(U^{\dagger}V)|^2 + d\right), \tag{A.3}$$

where V is the error unitary, and d is the dimension of the Hilbert space.
Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.2 is given by

$$\begin{aligned} F(V,U) &= \int d\psi \, \langle\psi| \, U^{\dagger}V \, |\psi\rangle \, \langle\psi| \, V^{\dagger}U \, |\psi\rangle \,, \\ &= \int d\psi | \, \langle\psi| \, W \, |\psi\rangle \, |^2, \end{aligned} \tag{A.4}$$

where $W = U^{\dagger}V$. Using the identity $\mathrm{Tr}(A)\mathrm{Tr}(B) = \mathrm{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V,U) = \mathrm{Tr}\left(\left(W \otimes W^{\dagger}\right) \int d\psi \left(|\psi\rangle \langle\psi|\right)^{\otimes 2}\right), \tag{A.5}$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitely rewritten as

$$\mathcal{M}_2 = \int d\psi \left(|\psi\rangle \langle\psi|\right)^{\otimes 2}. \tag{A.6}$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\mathrm{sym}} = \frac{1}{2}(\mathbb{I} + S). \tag{A.7}$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\mathrm{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\mathrm{Tr}(I) = d^2$ and $\mathrm{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)}\left(\mathbb{I} + S\right). \tag{A.8}$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \mathrm{Tr}\left[(W \otimes W^{\dagger})(\mathbb{I} + S)\right], \tag{A.9}$$

where we can evaluate the two terms using the identities $\mathrm{Tr}(A \otimes B) = \mathrm{Tr}(A)\mathrm{Tr}(B)$ and $\mathrm{Tr}((A \otimes B)S) = \mathrm{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter **??**. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.3.

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